



PhilaGive
The Philanthropy App



جامعة أم القرى
UMM AL-QURA UNIVERSITY

SOFTWARE ENGINEERING
PROJECT REPORT

PhilaGive

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An abstract graphic on the left side of the page, composed of numerous thin, red, curved lines that sweep upwards and then downwards, creating a sense of motion and depth. The lines are more densely packed in some areas, creating a darker red hue, while in others they are more sparse.

Planning Stage

Software Idea

PhilaGive is an application for collecting donations in which the user or the organization logs in, pays donations, and writes an article related to the case.

Application Type

This interactive transaction-based application needs a database to store information about users and organisations.

User Categories

Donors:

The app will enable donors to explore various charitable causes, make donations, and receive digital receipts

Charities:

Charities will have the ability to register their organizations, provide information about their missions and projects, and securely receive donations.

Requirements Specifications

Account Creation:

1. Allow users to create personal accounts using a username and a password.
2. Collect basic information during the account creation process, such as name, email, and contact details.

Registration:

1. Enable charities to register their organizations on the app.
2. Gather information about the charity, including name, mission, contact information, and proof of legitimacy.

Searching:

1. Provide a search functionality for users to find charitable causes and organizations.

Posting Articles:

1. Allow charities to post articles and updates about their ongoing projects, impact stories, and success stories.
2. Enable users to browse and engage with these articles, sharing them with others if desired.

Donations:

1. Enable donors to make monetary donations to registered charities through the app.
2. Support multiple payment options.
3. Ensure secure and encrypted transactions for donor peace of mind.
4. Provide digital receipts for donations made through the app for tax purposes.

Requests:

1. Allow charities to create requests for specific items or services they need for their projects.
2. Enable users to browse and respond to these requests, offering assistance or donations.



Analysis Stage

Functional Requirements for PhilaGive:

User and System requirement:

1.The user shall be able to create a new account.

1.1 system shall take their information from the registering form

(Username, password, full name, email, phone number, and location).

1.2 the system shall verify the user email and phone number if valid or not.

1.3 the system shall save the user information in the database

1.4 the system shall redirect the user to their profile page.

2.The user shall be able to login by username, phone number or email and password.

2.1 The system shall verify if the phone number or email address exist within the database.

2.2 the system shall log in the user and direct them to homepage

Functional Requirements for PhilaGive:

User and System requirement:

3. Provide a search functionality for users to find charitable causes and organizations.

3.1 The search functionality shall provide users with options to search by keywords, categories, location, or other relevant criteria.

3.2 The system shall retrieve and display search results based on the user's search query and selected filters.

3.3 The search results shall present relevant information about the charitable causes and organizations, (including names, descriptions, contact information, and any associated media).

3.4 The system shall provide sorting and filtering options to refine search results based on user preferences.

Functional Requirements for PhilaGive:

User and System requirement:

4.Allow charities to post articles and update about their ongoing projects, impact stories, and success stories.

4.1 The system shall provide a feature that allows registered charities to create and publish articles and updates.

4.2 The article creation feature shall include options for charities to add titles, descriptions, images, and relevant content.

4.3 The system shall provide a user-friendly editor for charities to format and customize the articles.

4.4 Charities shall have the capability to edit, update, or remove their published articles as needed.

5.Enable users to browse and engage with these articles, sharing them with others if desired.

5.1 The system shall provide a user interface that allows users to browse and search for articles published by charities.

Functional Requirements for PhilaGive:

User and System requirement:

5.2 The user interface shall display articles in an organized and visually appealing manner, including titles, summaries, and featured images.

5.3 Users shall have the ability to click on an article to view its full content, including images and embedded media.

5.4 The system shall provide options for users to interact with articles, such as liking, commenting, and sharing them.

6.Enable donors to make monetary donations to registered charities through the app.

6.1 The system shall provide a secure and user-friendly interface for donors to make monetary donations.

6.2 The donation feature shall support multiple payment options, such as credit cards, debit cards, and digital wallets.

6.3 The system shall ensure the security and encryption of donor's payment information and transaction details.

Non-functional Requirements for PhilaGive:

1. Usability:

- The app should have an intuitive and user-friendly interface.
- The app should provide clear instructions and guidance to users.
- The app should be responsive and provide a smooth user experience.
- The training time for users to become proficient in using the app should be less than 2 hours .

2. Performance:

- The app should have fast response times, ensuring minimal delays during user interactions.
- The app should be able to handle a large number of concurrent users without performance degradation.
- The app's response time should be within acceptable limits, typically responding to user actions within 2 seconds.
- The app should have a high rate of responsiveness, with a goal of processing user requests at a rate of at least 100 requests per second.

3. Security:

- User data, including personal and financial information, should be securely stored and transmitted.
- Secure authentication mechanisms, such as encryption and secure protocols, should be implemented.
- The app should be protected against common security threats, such as unauthorized access and data

Non-functional Requirements for PhilaGive:

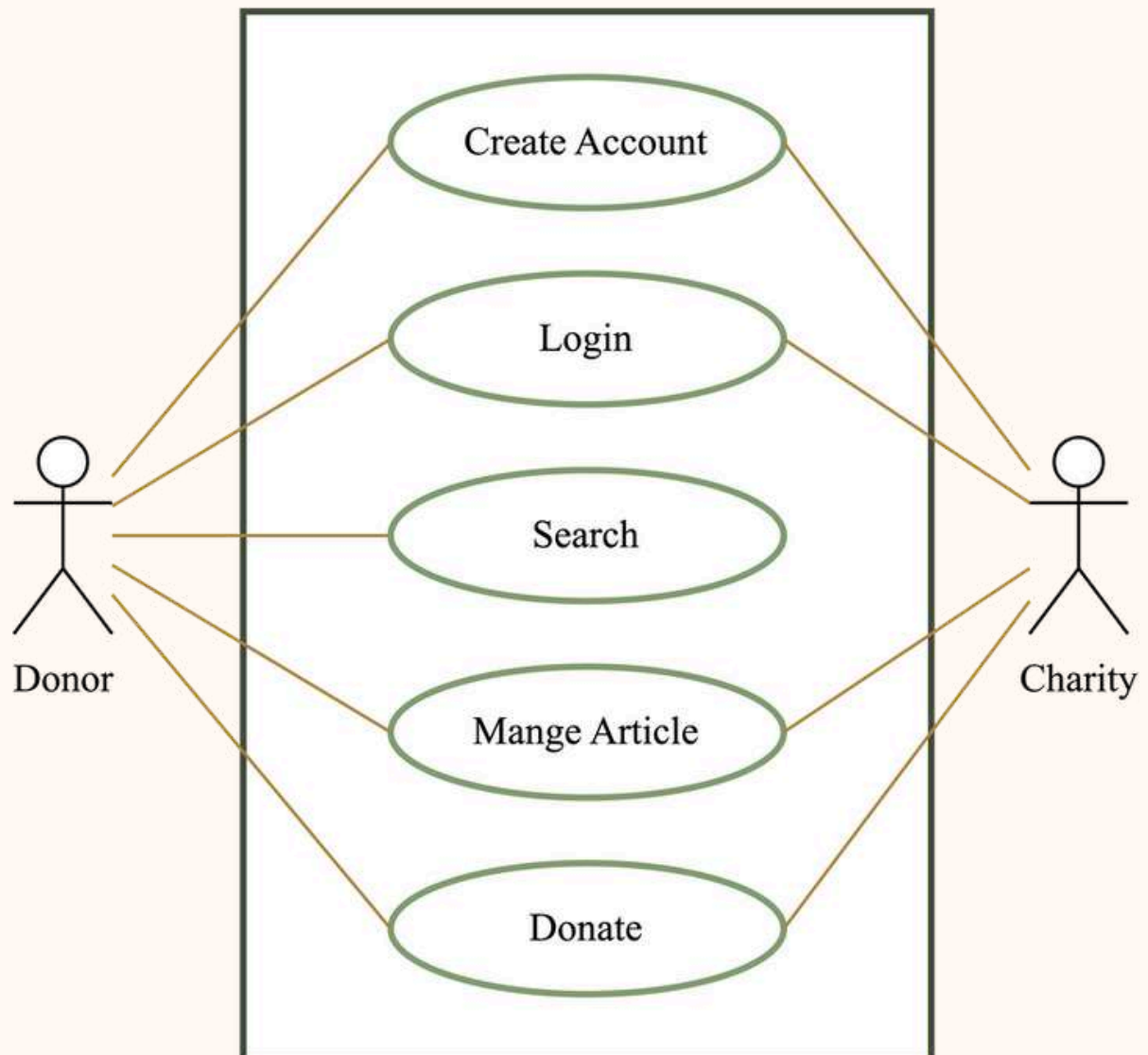
4. Reliability:

- The app should be available and accessible to users without frequent downtime or disruptions.
- The app should handle errors and exceptions gracefully, providing appropriate error messages and recovery options.
- The app's failure occurrence rate should be minimal, with a goal of less than 0.5% of failures per second.

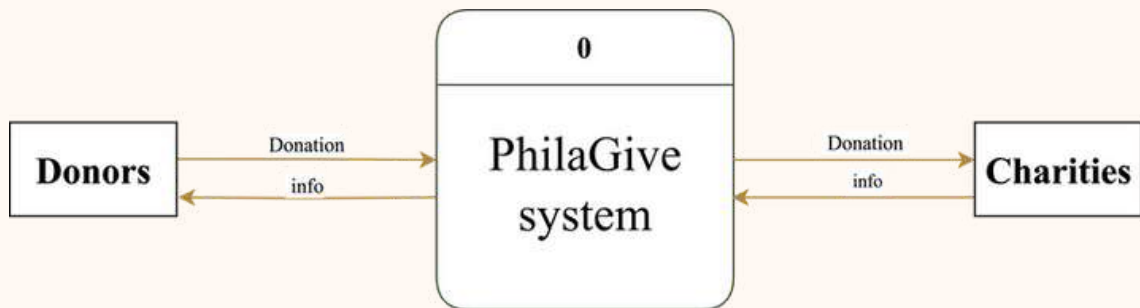
5. Scalability:

- The app should be designed to accommodate future growth in user numbers and data volume.
- The app's architecture should support scalability and be able to handle increased user demand.

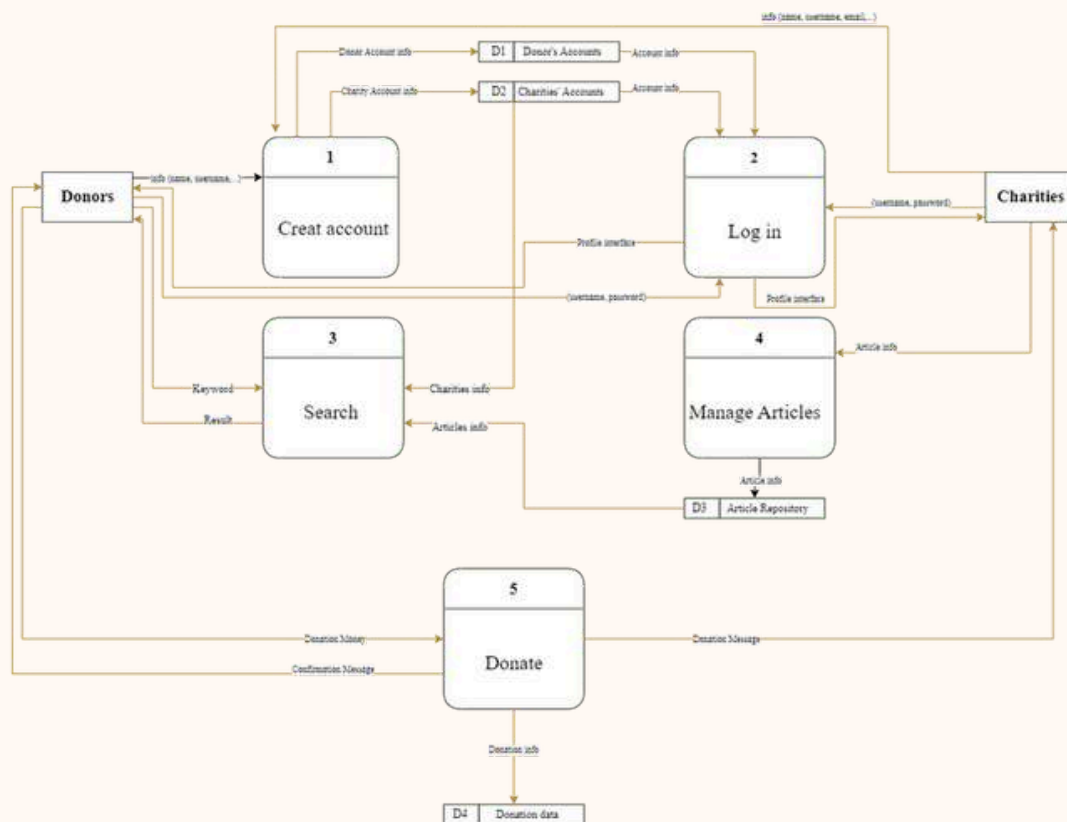
Use Case Diagram



Context Diagram

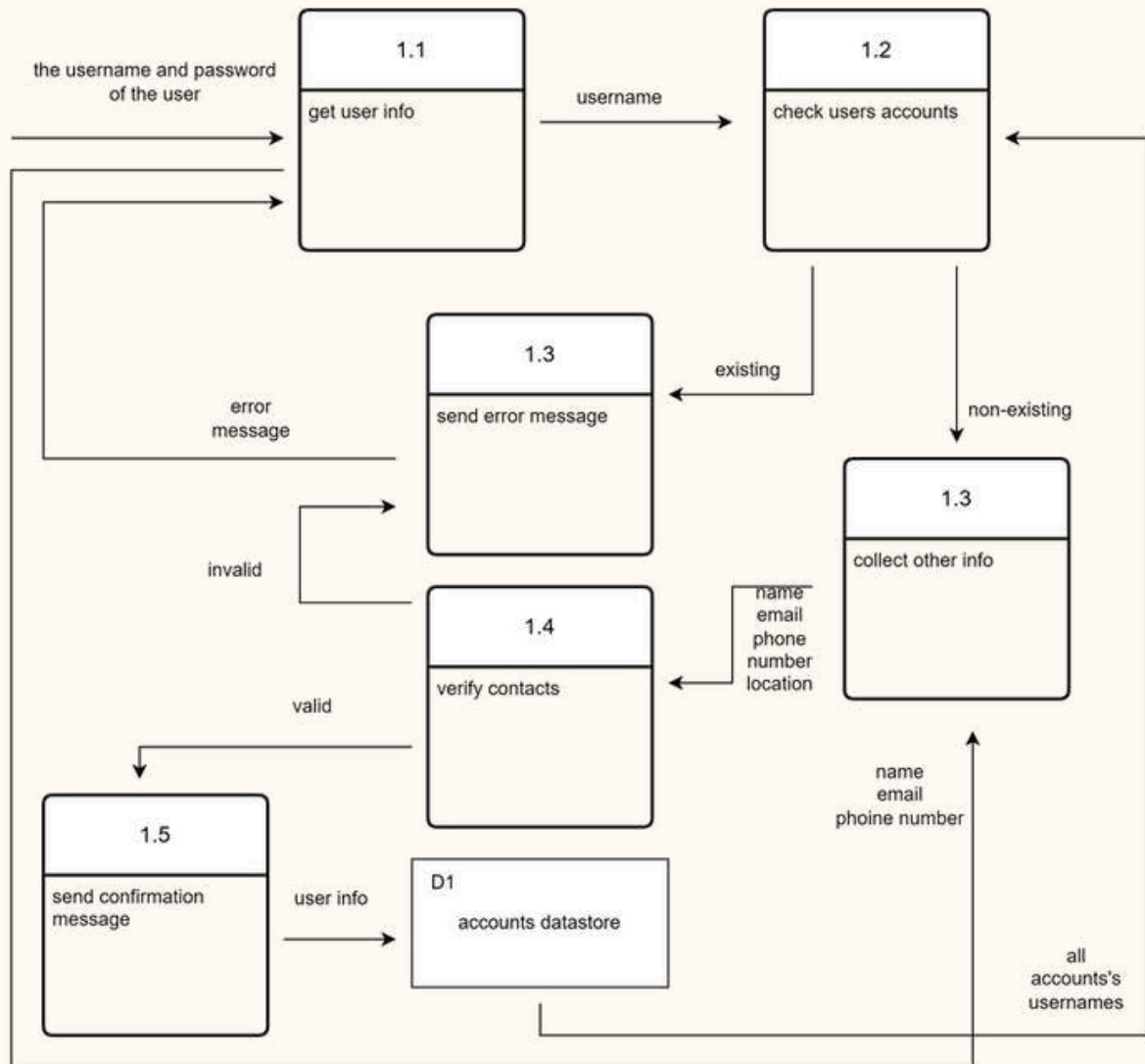


DFD Level1

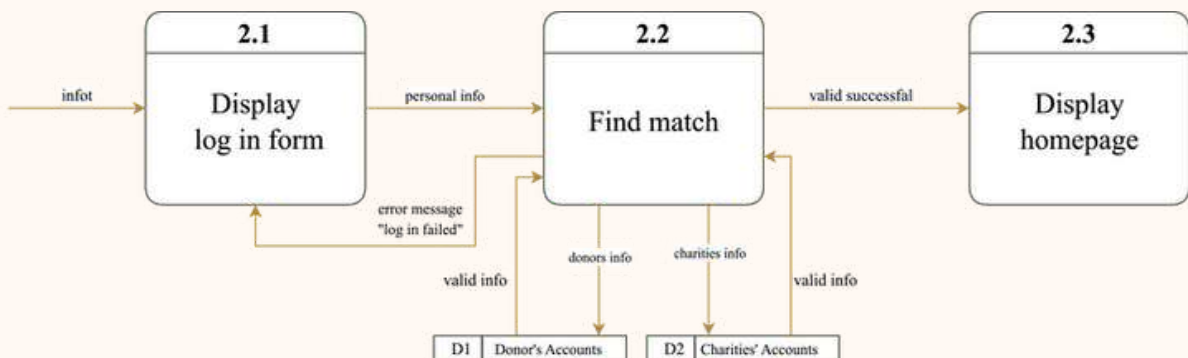


Child Diagram

Child level “Account creation” :

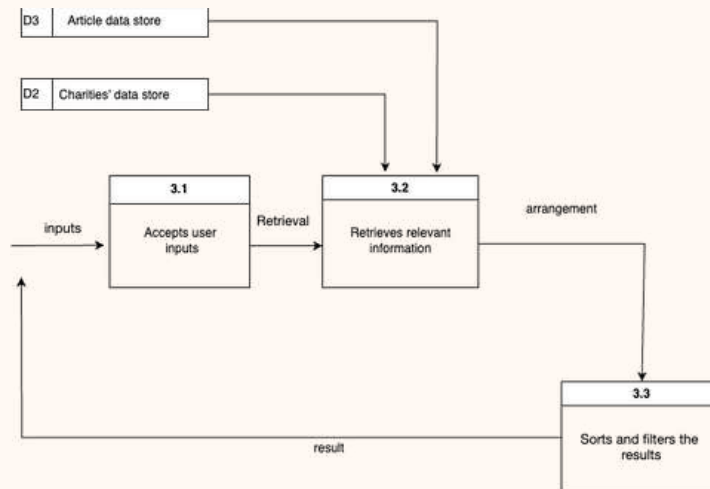


Child level “Log in” :

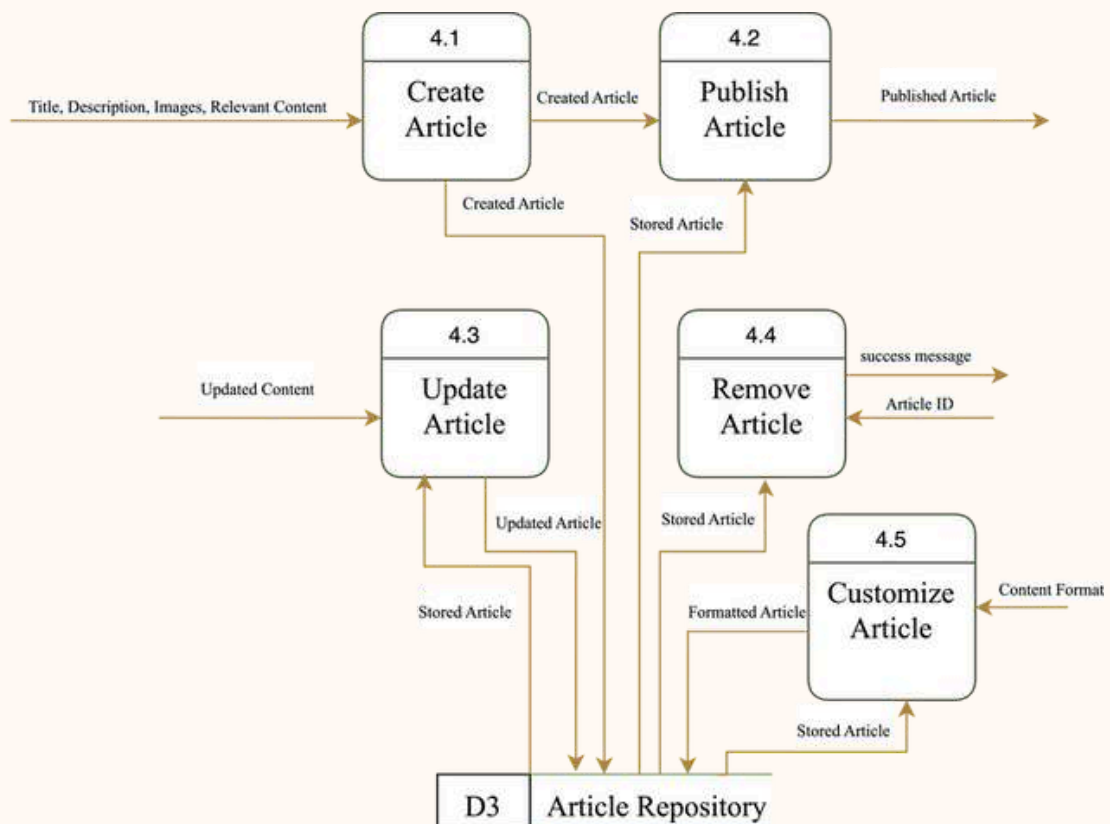


Child Diagram

Child level “Search” :

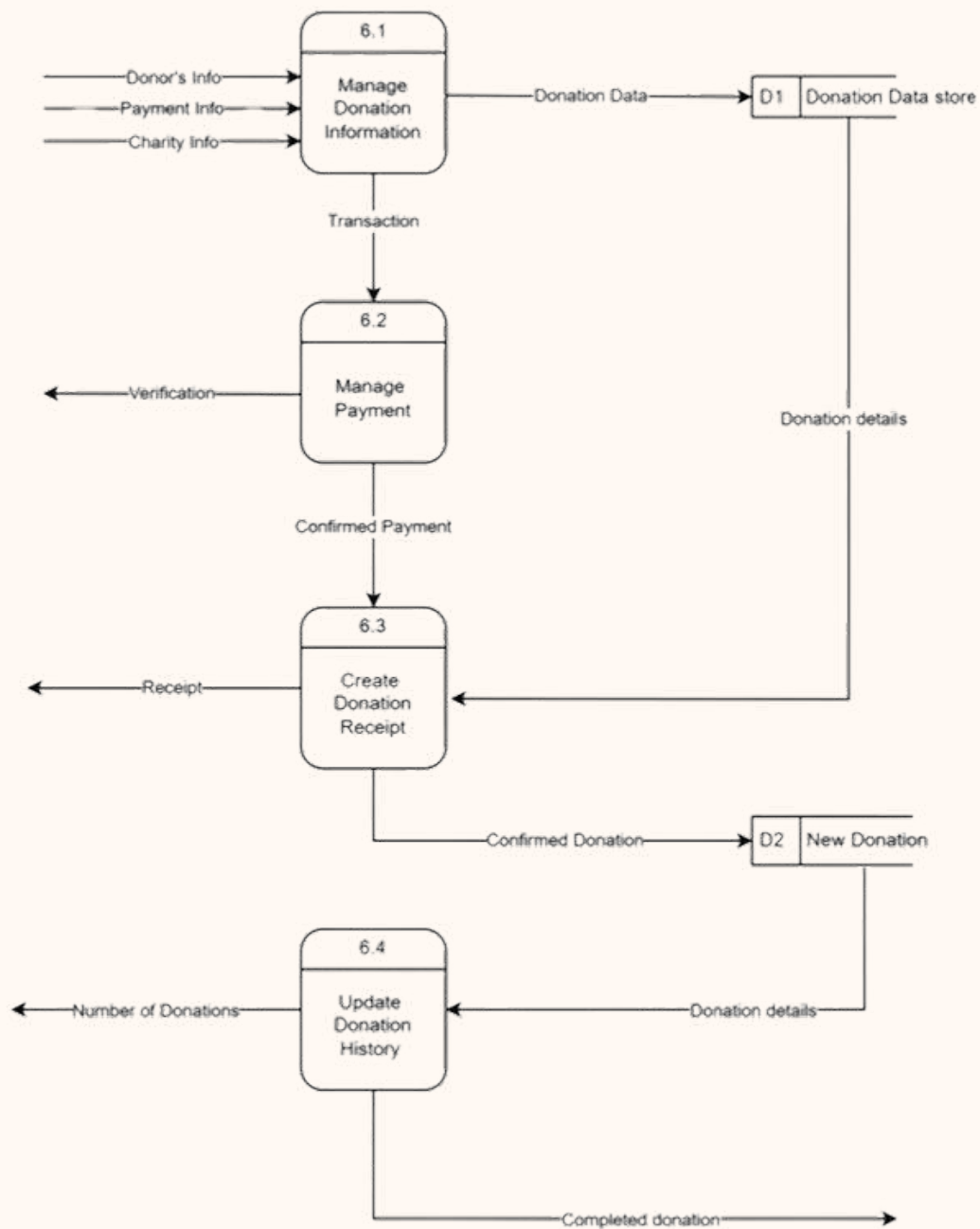


Child level “Manage Articles” :



Child Diagram

Child level “Donation” :



scenarios

account creation scenarios

Actor	the user (Donors and Charities)
Description	The user will provides a username and password for account creation. The system checks if there's an existing user with the same username. If the username is unique, the system requests additional information such as name, phone number,location and email. The system verifies the provided information for validity. If the information is valid, it is stored in the datastore as a new account and a confirming message will be displayed. If the information is invalid, an error message is displayed to inform of the issue. validation of provided information, such as email and phone numbers, will help maintain data accuracy and integrity in the database.
Data	collected data : (username , passwords , name , phone number, email,location)
Stimulus	there will be a button called (submit)
Responses	confirming message (your account created successfully)
Comments	after verification the user will be able to use their accounts. administrators can view the user info except passwords and usernames. passwords and usernames will be checked automatically by the system

scenarios

Log in

Actor	System user
Description	The user shall be able to login by username, phone number or email and password. The system shall verify if the phone number or email address exist within the database. The system shall log in the user and direct them to homepage.
Data	User personal information .
Stimulus	Log in button .
Responses	Redirect the users to the homepage.
Comments	non

scenarios

Manage Articles

Actor	Charity
Description	1. Allow charities to post articles and update about their ongoing projects, impact stories, and success stories. 1.1 The system shall provide a feature that allows registered charities to create and publish articles and updates. 1.2 The article creation feature shall include options for charities to add titles, descriptions, images, and relevant content. 1.3 The system shall provide a user-friendly editor for charities to format and customize the articles. 1.4 Charities shall have the capability to edit, update, or remove their published articles as needed.
Data	Article title, description, images, content
Stimulus	The user will click on the required button
Responses	Displays a confirmation message
Comments	The system may perform validations on the data provided by the charity before publishing the article.

scenarios

Search

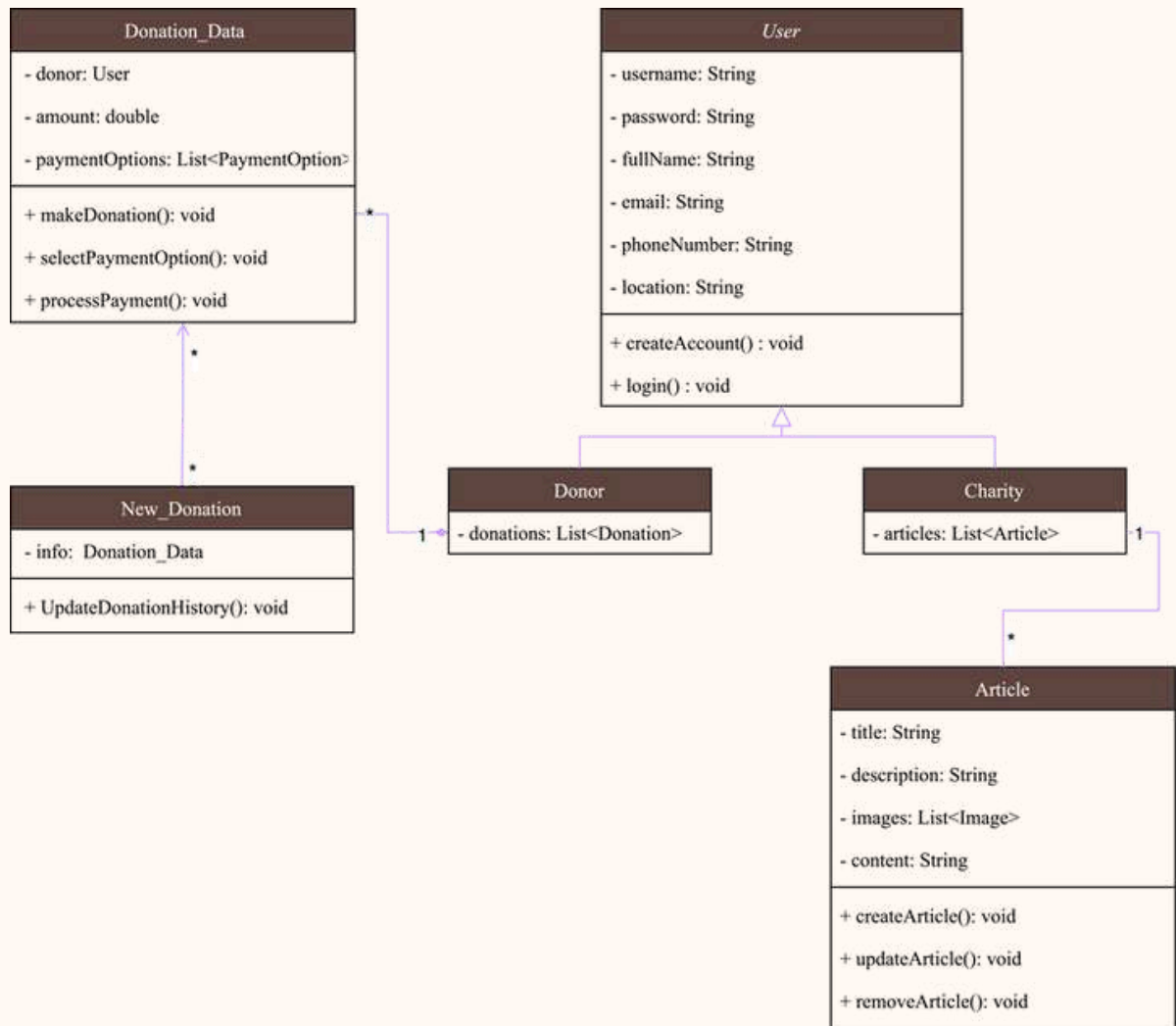
Actor	users
Description	3.1 The search functionality shall provide users with options to search by keywords, categories, location, or other relevant criteria about the Charity and articles 3.2 The system shall retrieve and display search results based on the user's search query and selected filters. 3.3 The search results shall present relevant information about the charitable causes and organizations,(including names, descriptions, contact information, and any associated media). 3.4 The system shall provide sorting and filtering options to refine search results based on user preferences.
Data	user's inputs (keywords, categories, location)
Stimulus	button to send the research
Responses	result for the user
Comments	non

scenarios

Donation Scenario

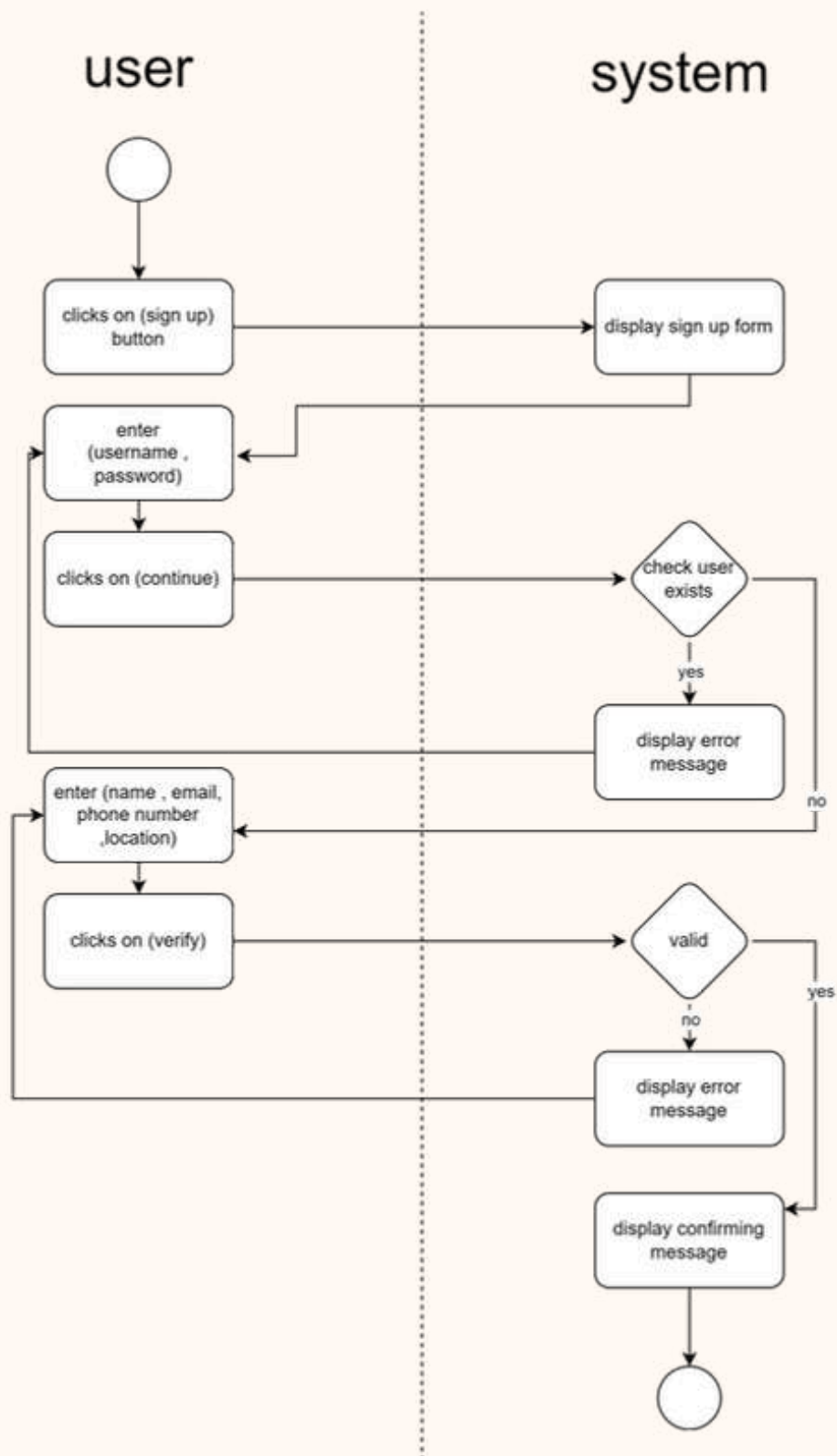
Actor	Donor and Charity
Description	enables registered donors to make monetary donations to registered charities through the app. It also allows donors to view their donation history, providing them with a comprehensive record of their past contributions. prompting the donor to enter the donation amount and select a payment option. confirm the donor, acknowledging the successful donation and providing relevant details. The system retrieves and displays the donor's donation history, including information such as donation amounts, dates, and recipient details.
Data	• Donor's Info(registered info). • Payment info(amount, selected payment option and their required details such as: card number, expiration date, CVV). • Charity Info(charity name). • Confirmation messages, Number of Donations, Receipts
Stimulus	• "Donate" and "View Donation History" Buttons • Donor enters the desired donation amount and selects the preferred payment option.
Responses	sends a confirmation message and recipient details.
Comments	The donation feature should comply with relevant security and encryption standards to ensure the protection of donor's payment information.

Class Diagram



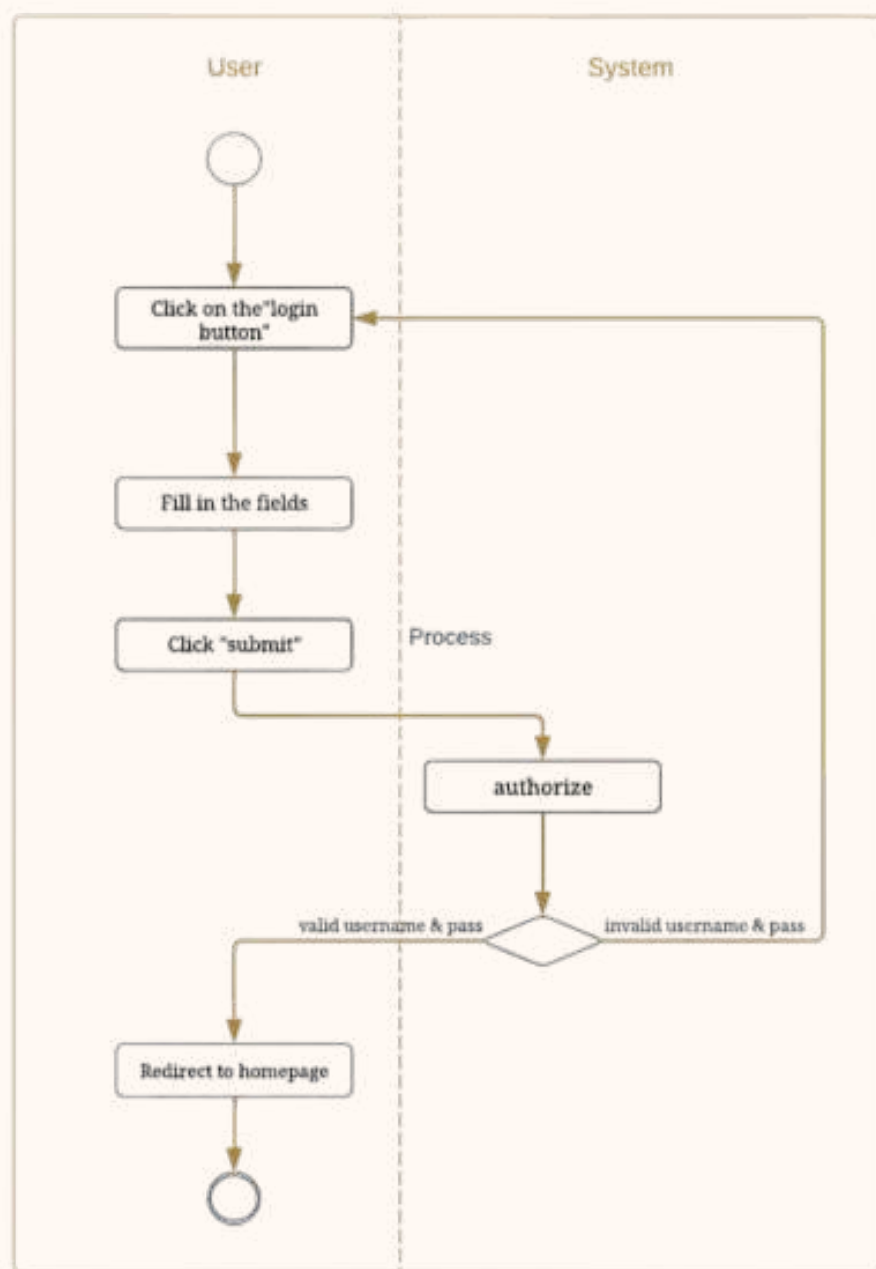
Activity Diagrams

[account creation]



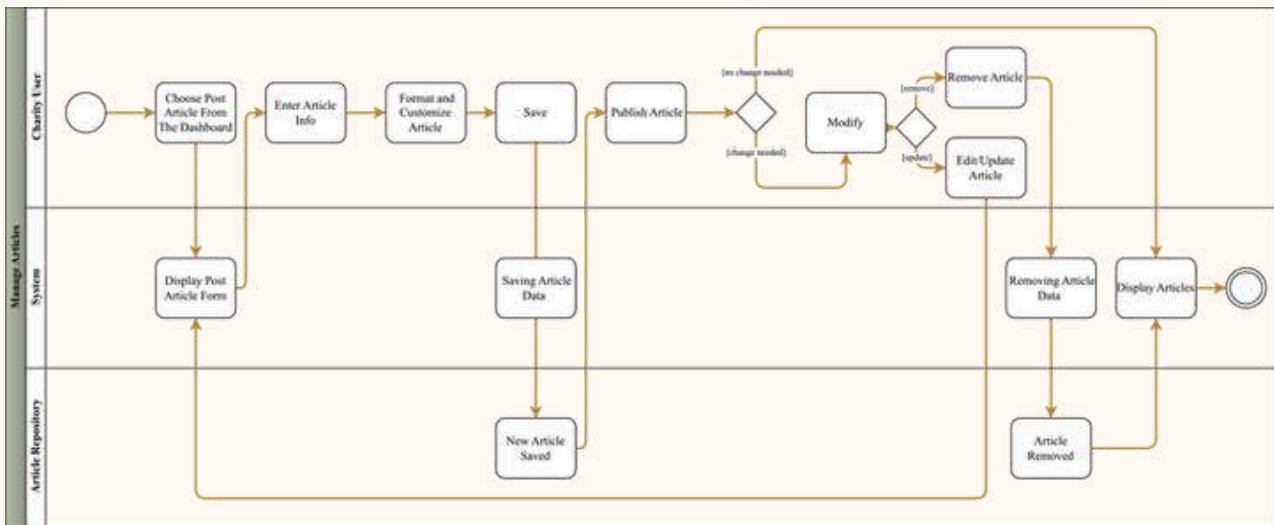
Activity Diagrams

[Login] :



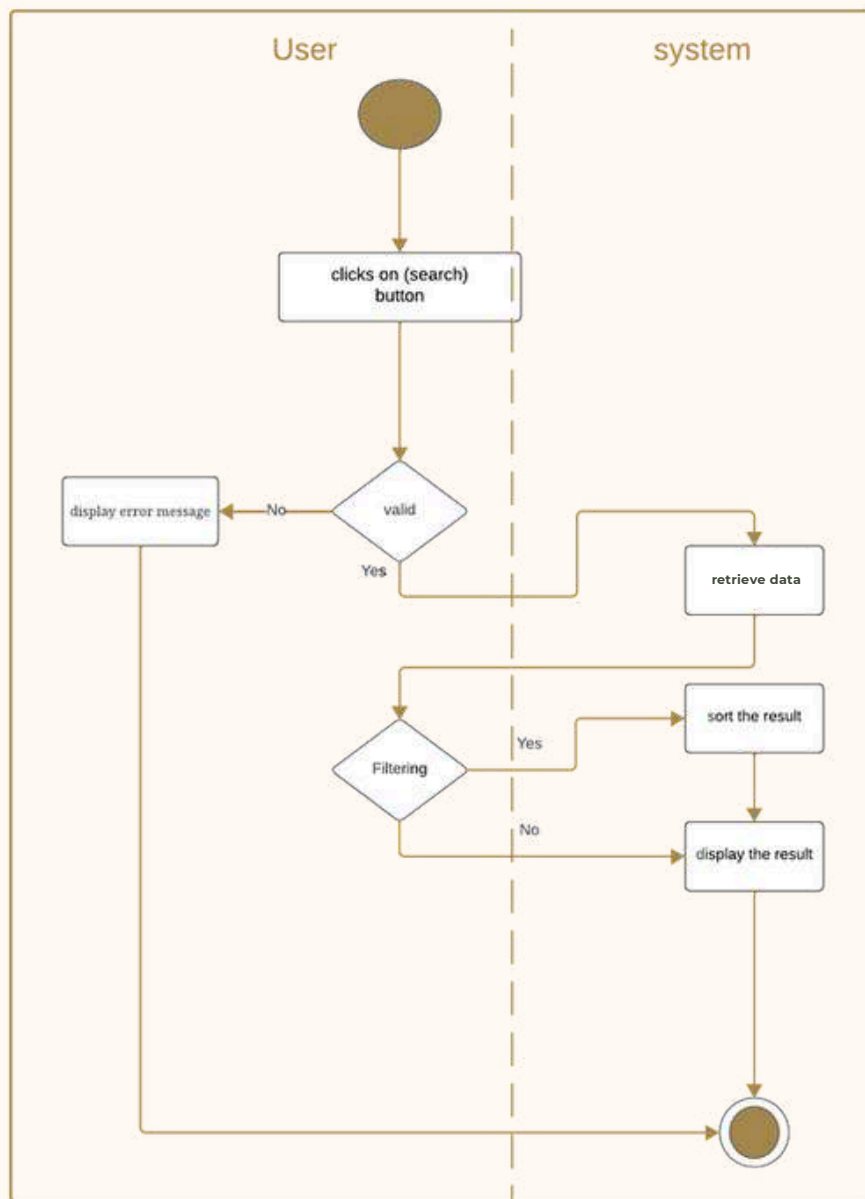
Activity Diagrams

[Manage Articles]:



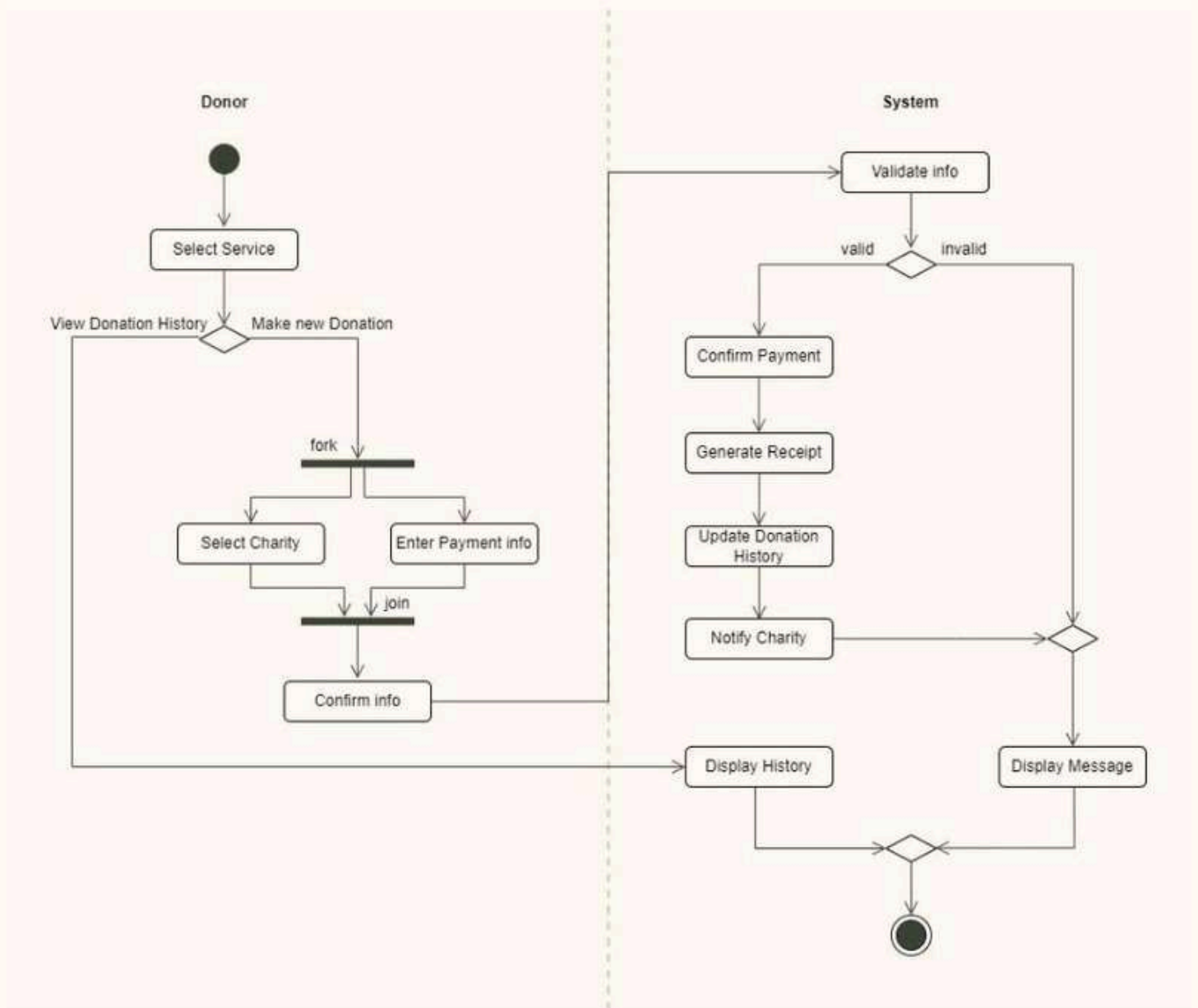
Activity Diagrams

Search



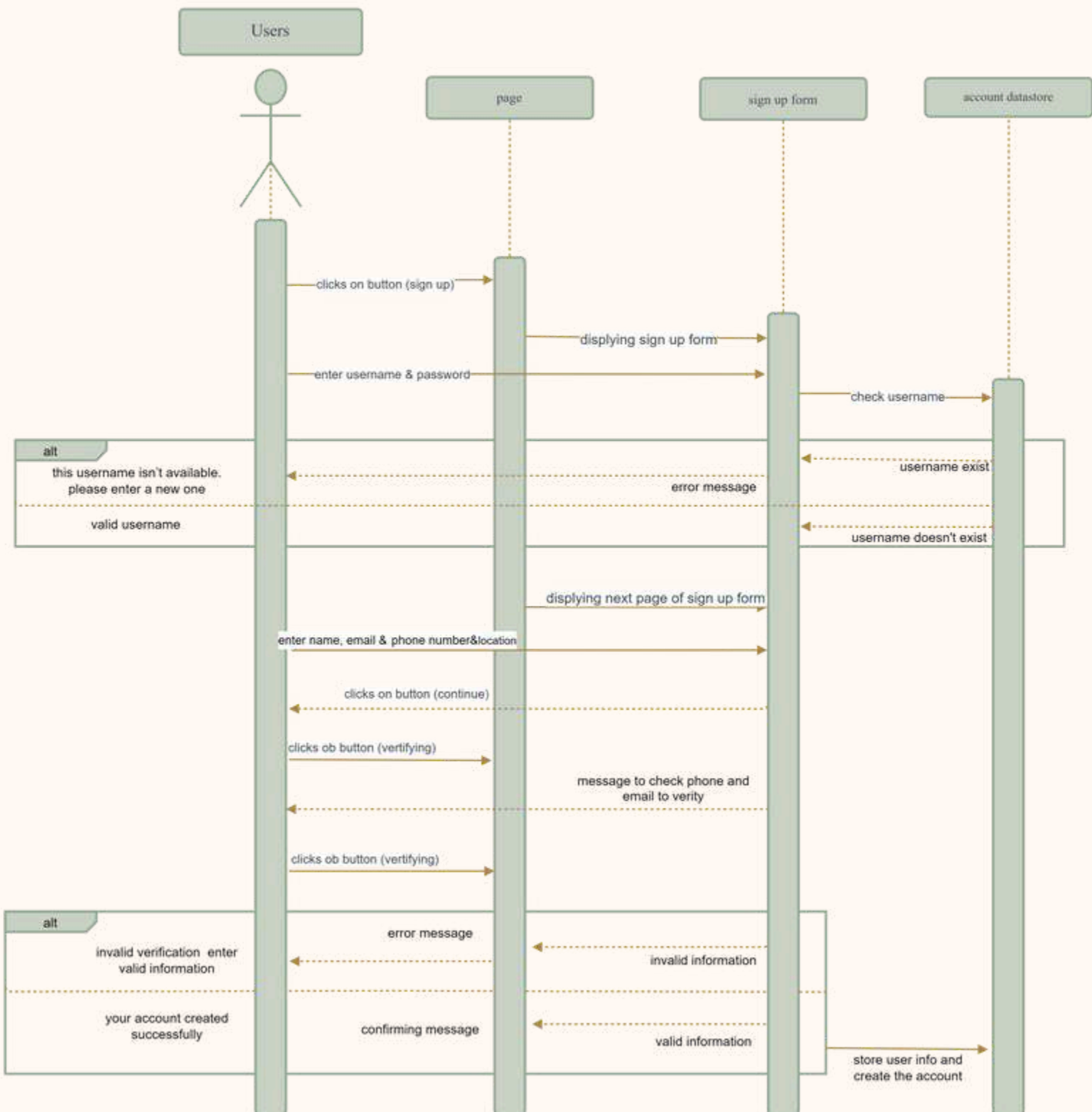
Activity Diagrams

[Donate]



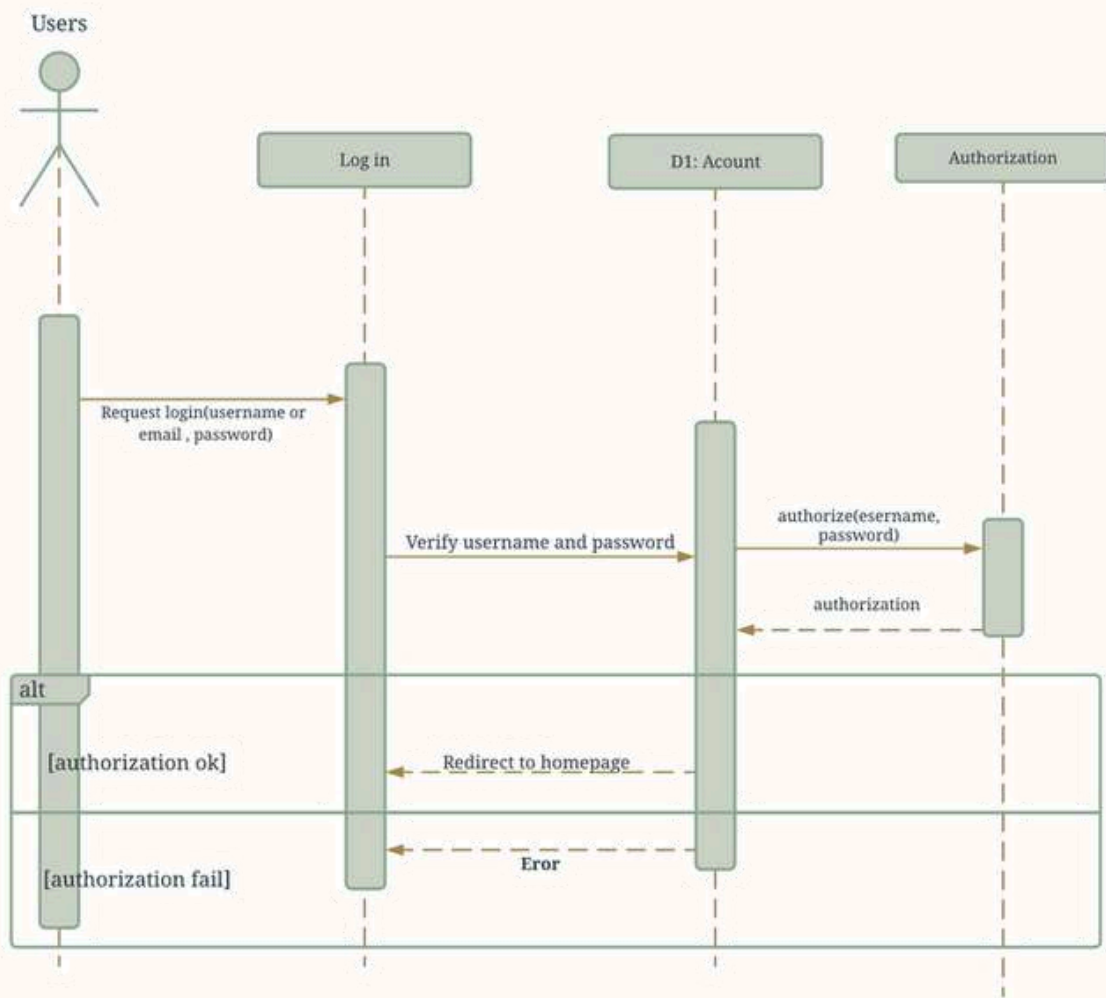
Sequence Diagrams

[Account Creation]:



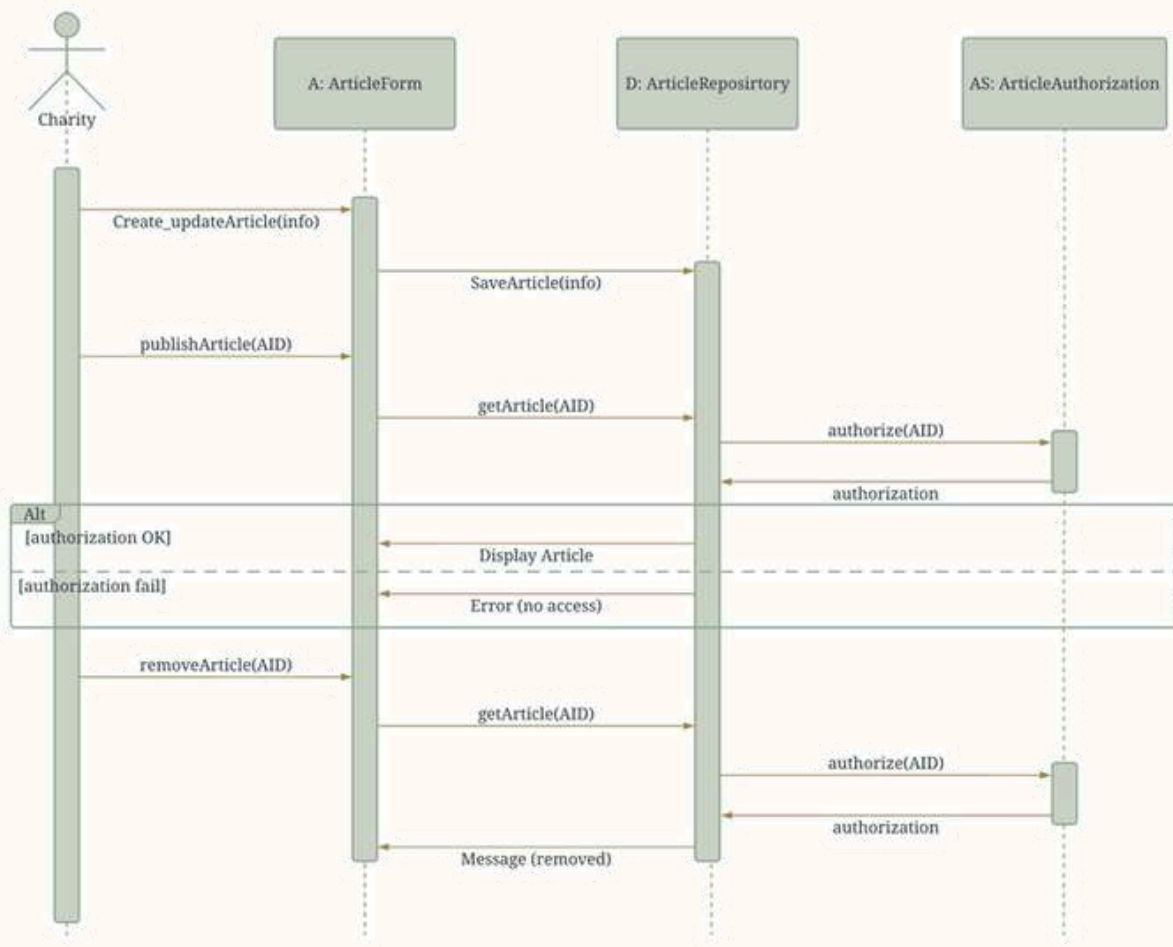
Sequence Diagrams

[Login] :



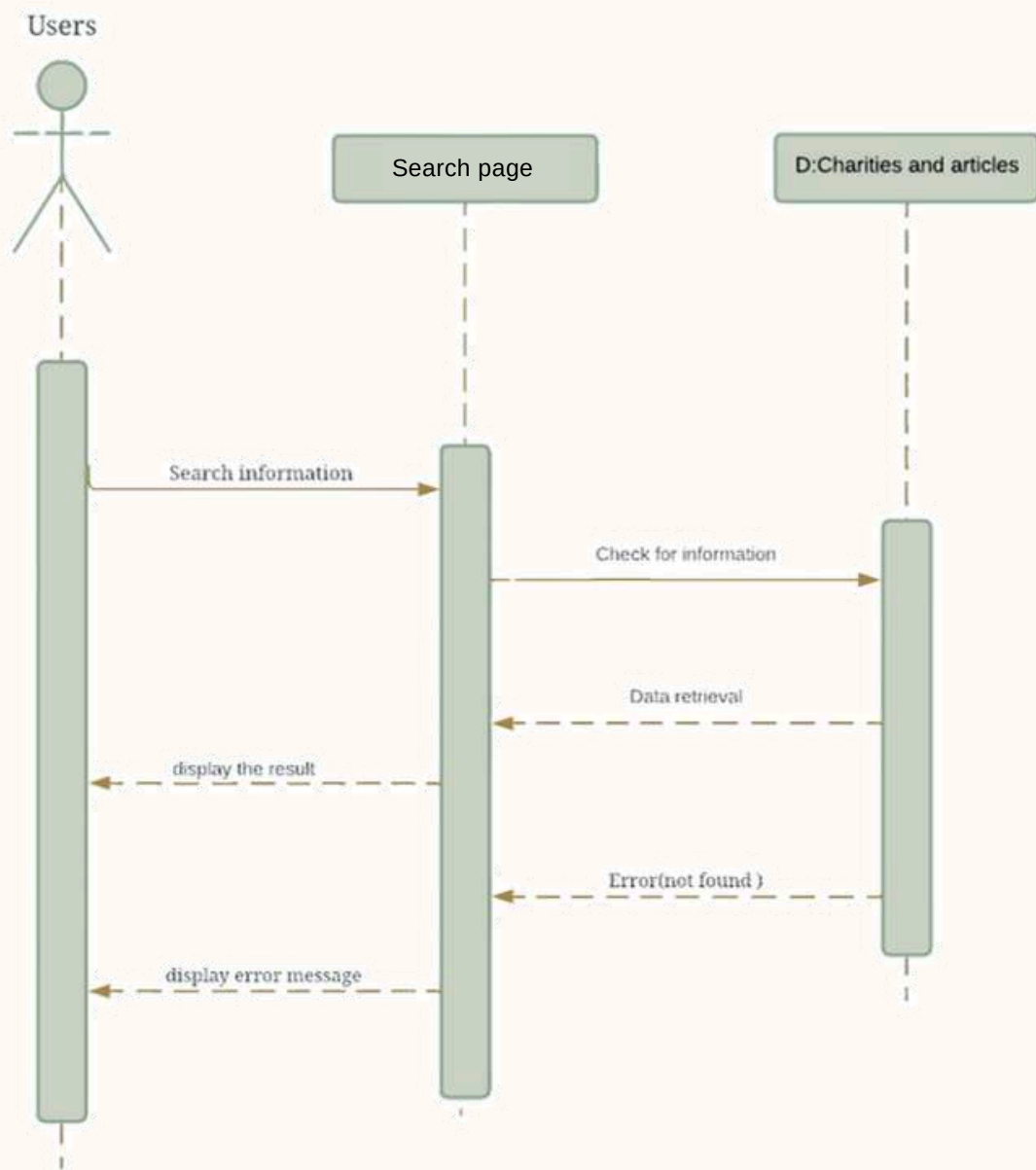
Sequence Diagrams

[Manage Articles] :



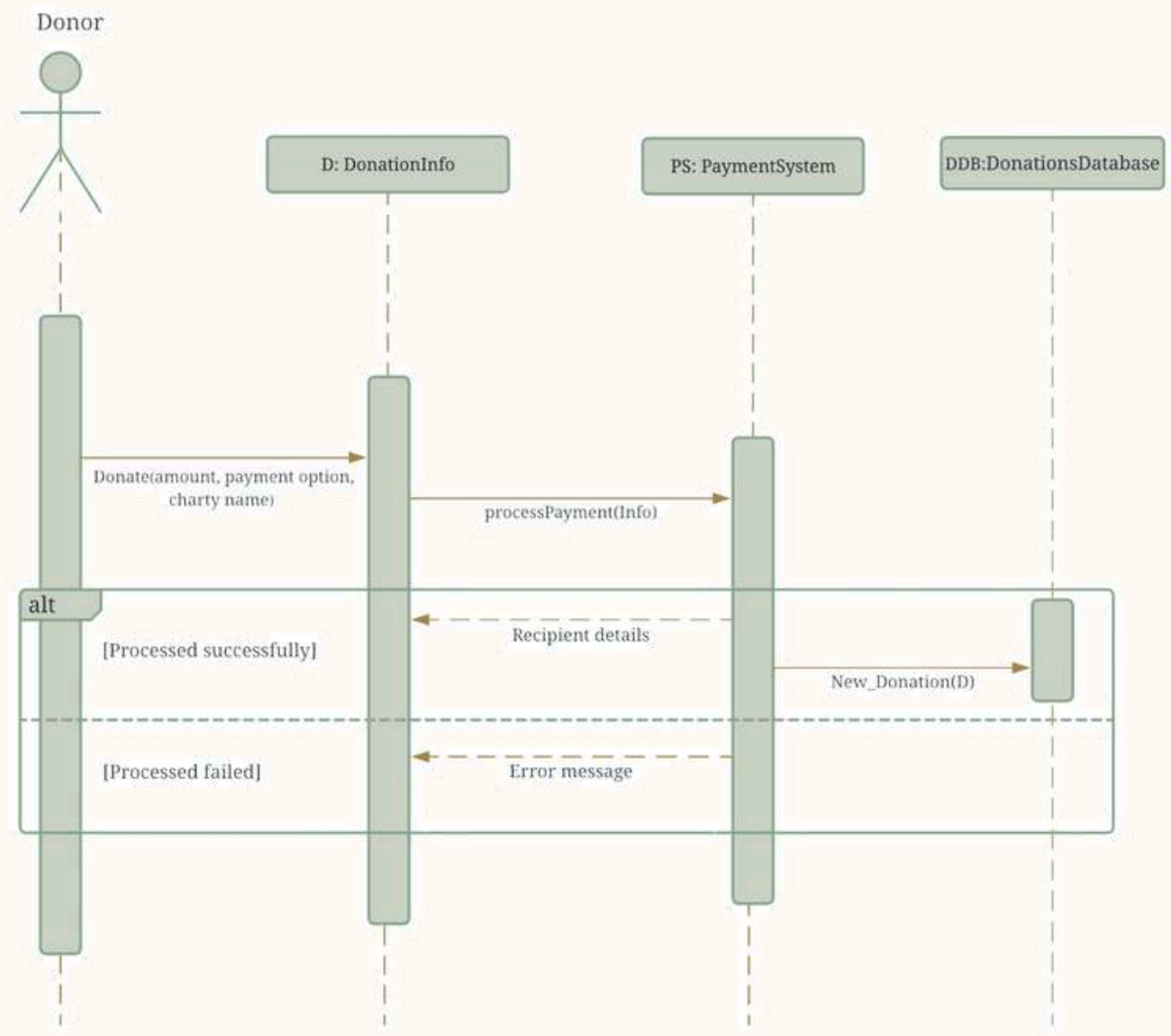
Sequence Diagrams

[Search]



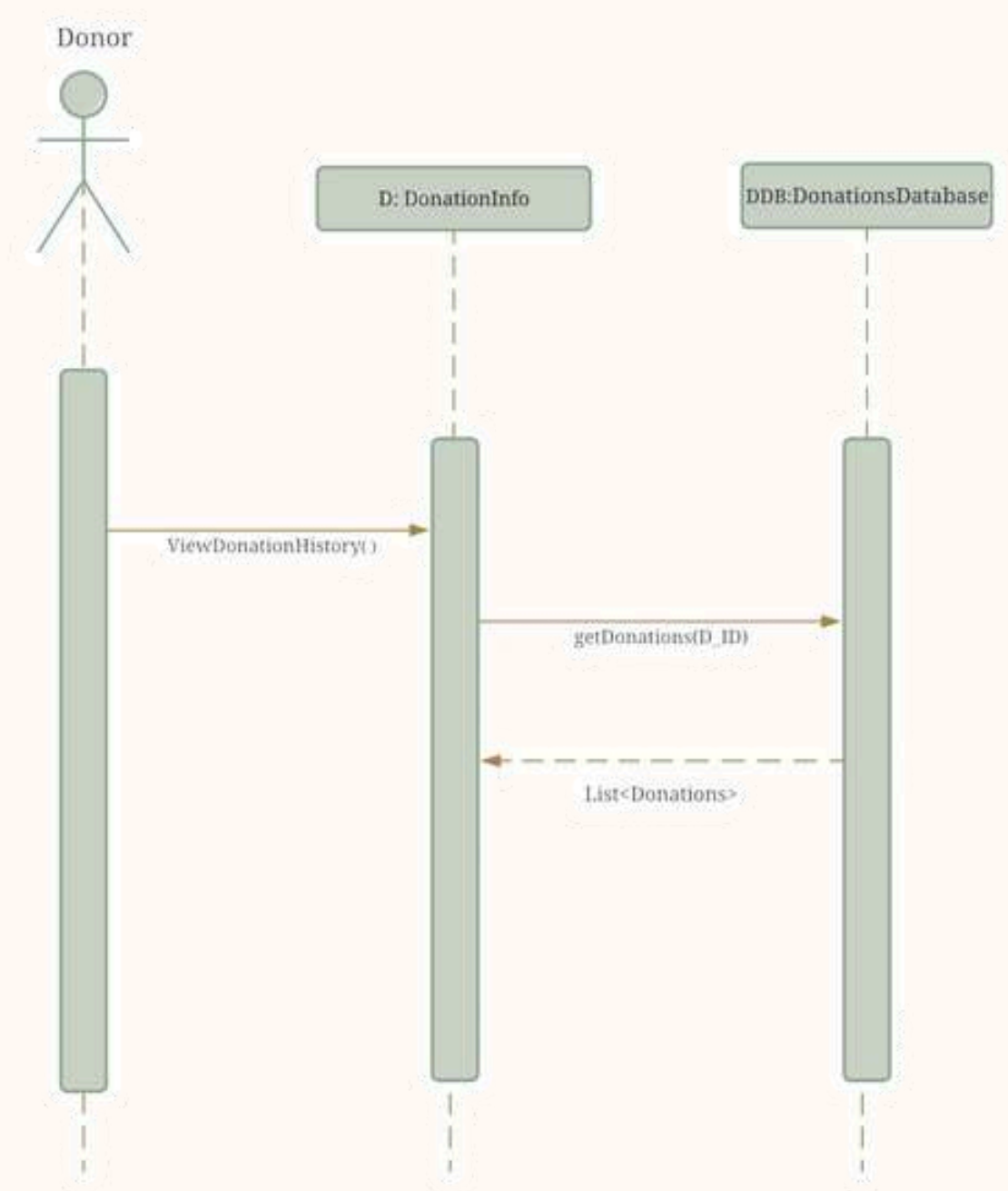
Sequence Diagrams

[Donate]:

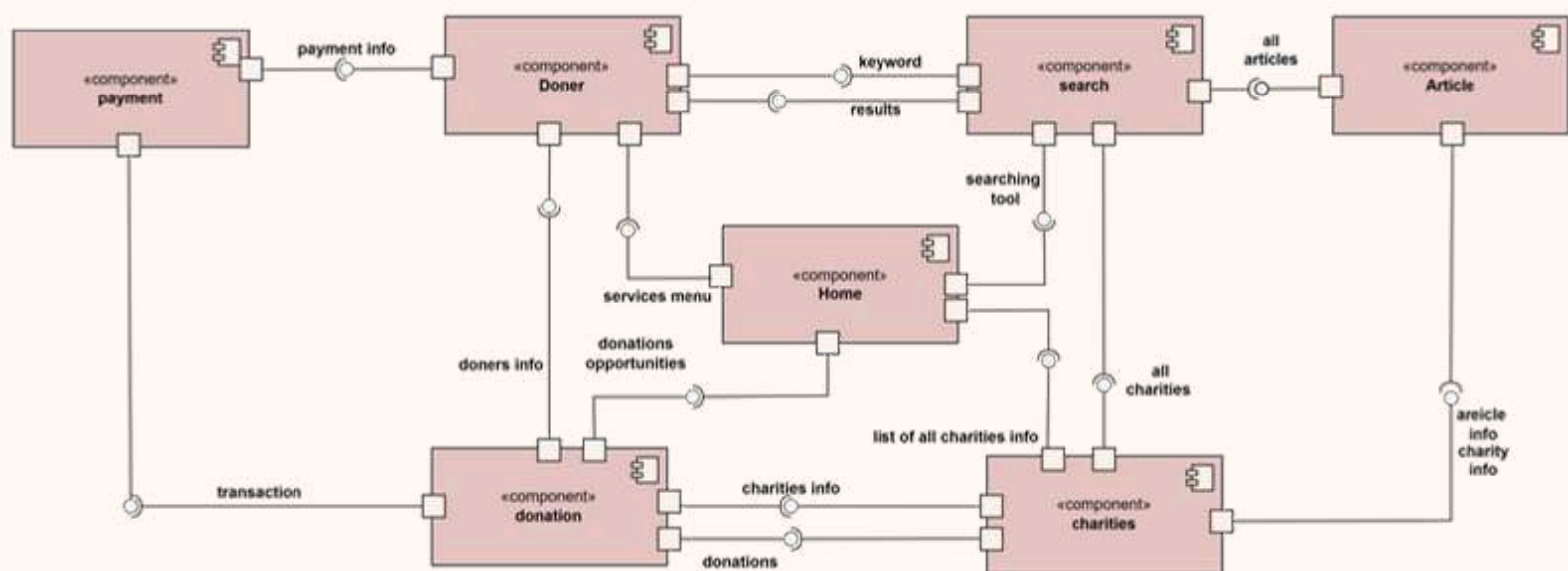


Sequence Diagrams

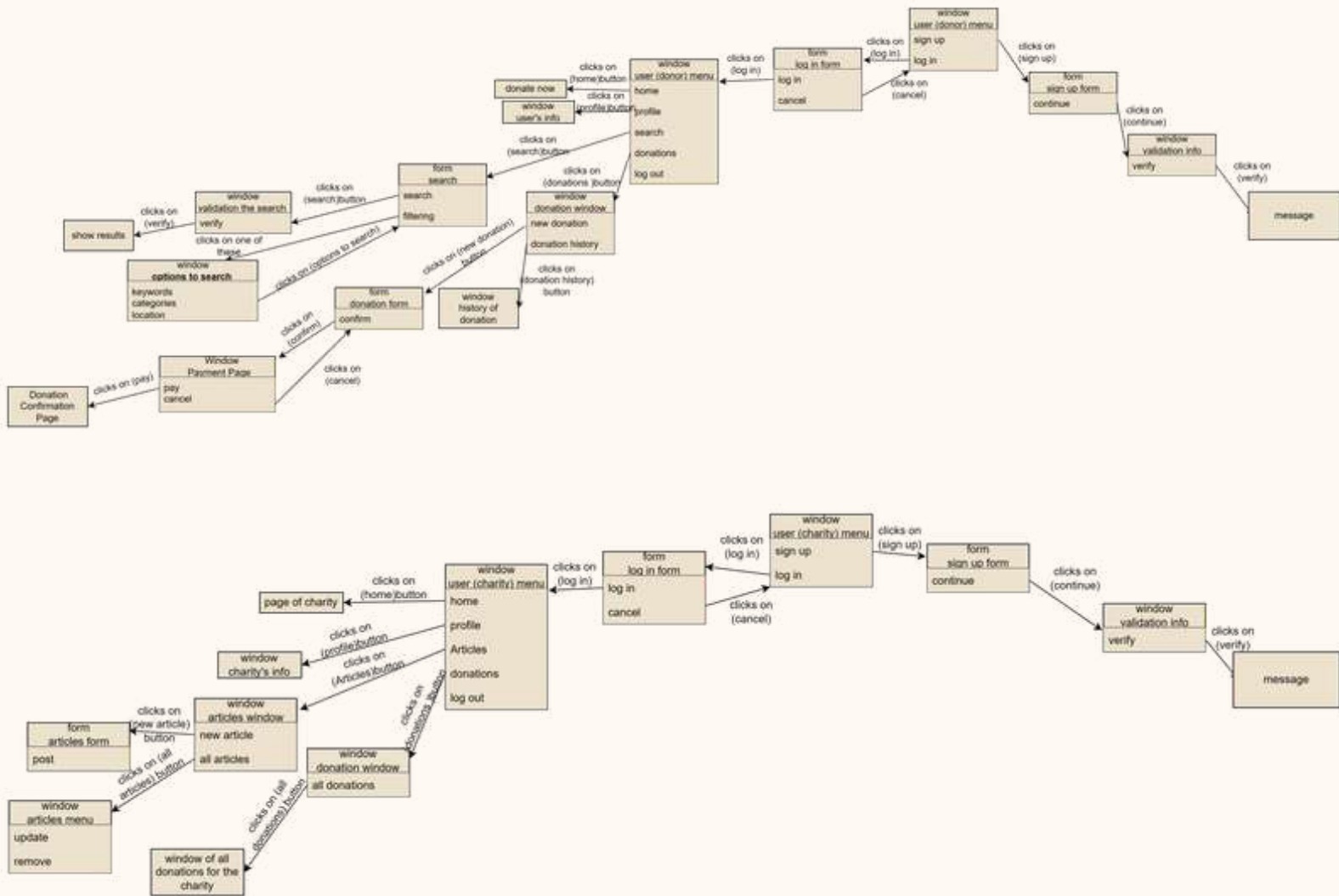
[View Donation History]:



Component Design

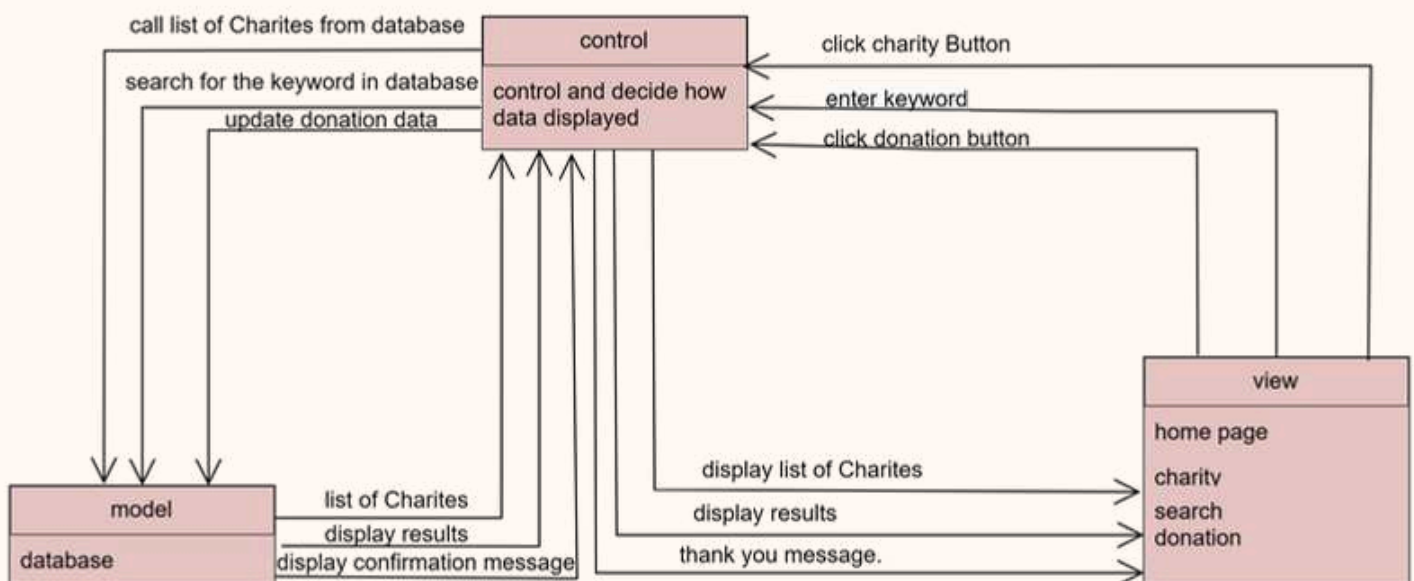


Interface Design

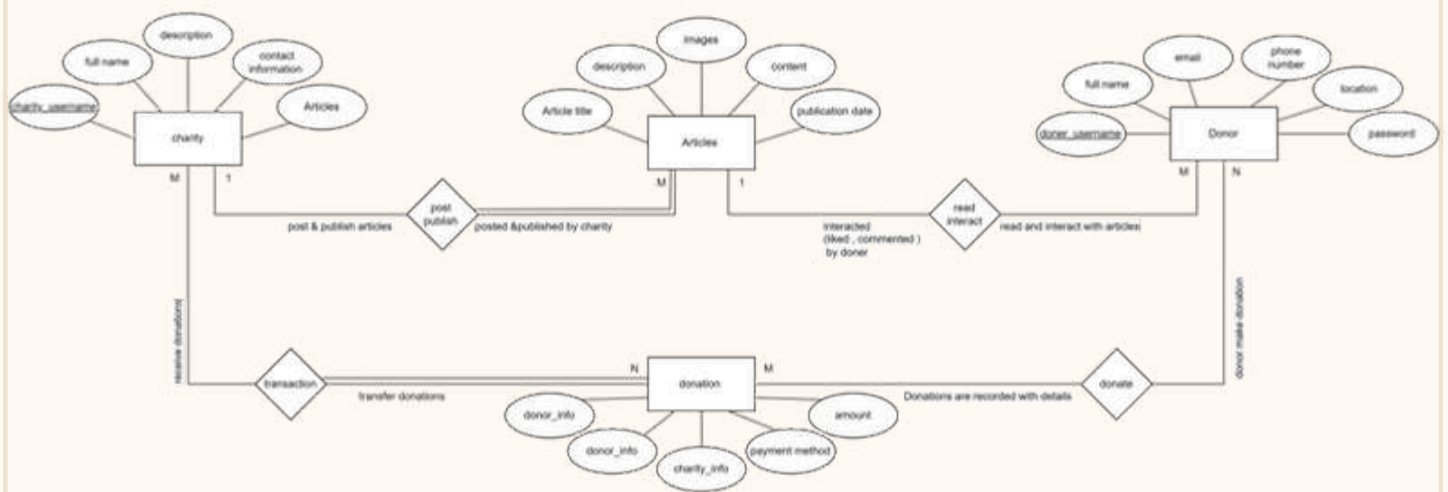


Architectural design

MVC architectural design pattern is well-suited for the PhilaGive donation application due to its clear separation of concerns, modularity, reusability, scalability, user interface flexibility, maintainability, and support for multiple platforms, all of which are crucial aspects for the successful development and operation of PhilaGive.



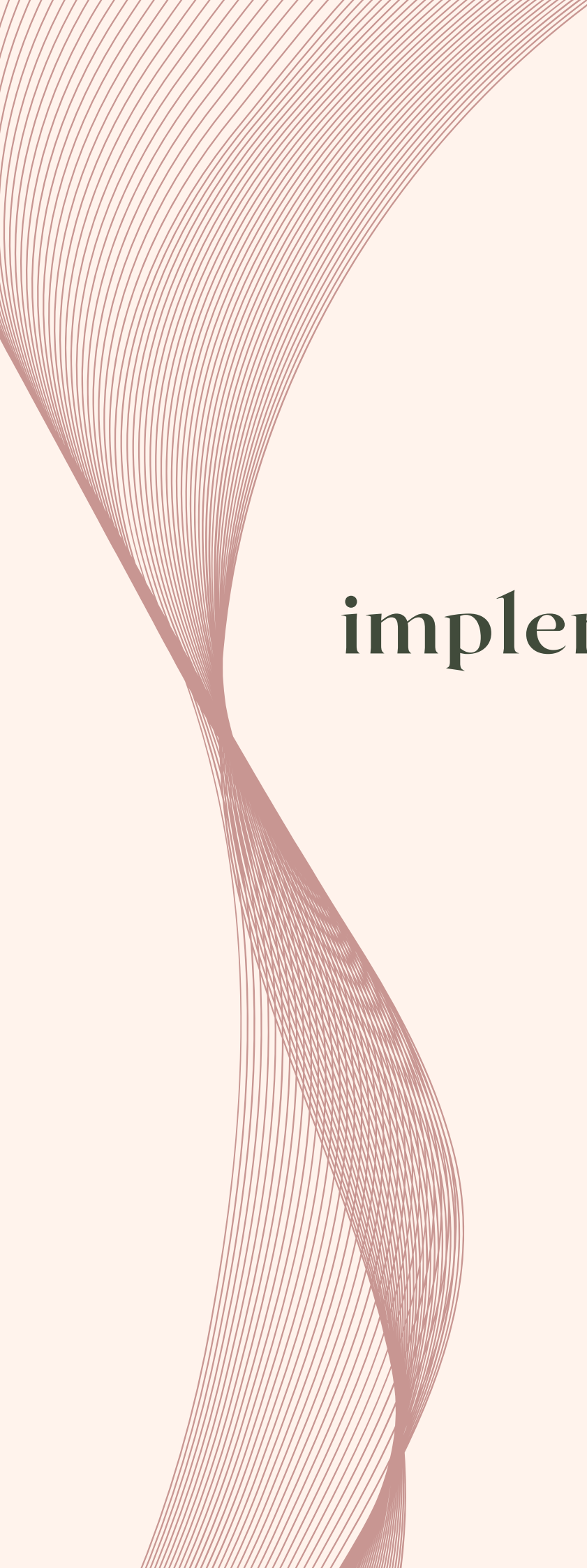
Entity Relationship Diagram



Optimization Techniques:

we apply indexing to the primary and foreign keys used in the relationships between entities. Specifically:

- Indexing 'charity_id' in the 'Articles' entity to enhance retrieval of articles associated with a particular charity efficiently.
- Indexing 'donor_id' in the 'Donation' entity to speed up retrieval of donation information linked to a specific donor.
- Implementing database normalization techniques to eliminate redundant data and ensure data integrity, thereby optimizing storage and retrieval processes.

The image features a series of thin, red, wavy lines on the left side, creating a sense of movement and depth. These lines curve and flow from the top left towards the bottom left, with some lines crossing each other. The overall effect is a dynamic, organic shape that contrasts with the solid light beige background.

implementation

all members

welcome page

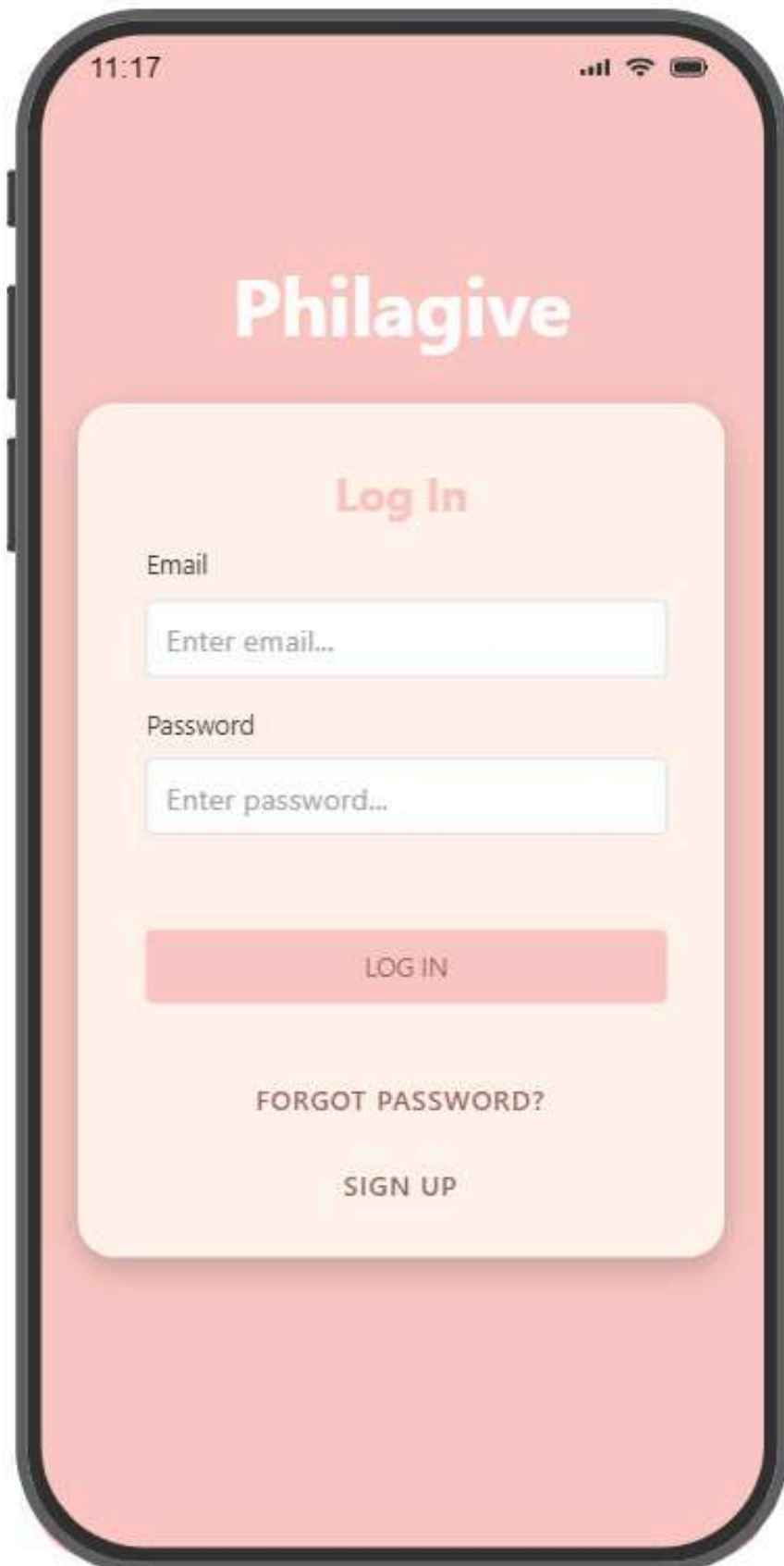
the first page of the app



log in page

futun

after pressing (already have an account)



11:17

Philagive

Log In

Email

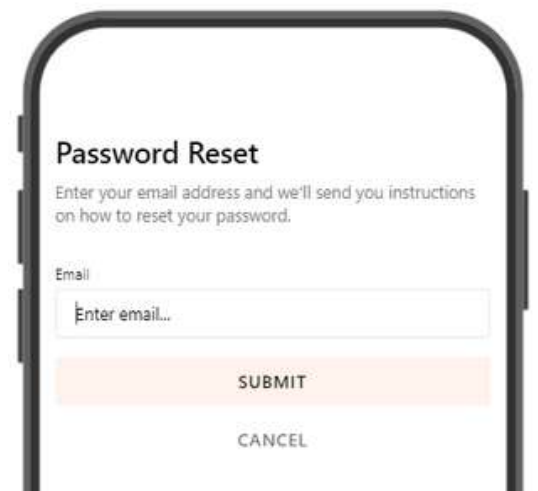
Password

LOG IN

FORGOT PASSWORD?

SIGN UP

pressing on
(forget password)
will move to the
reset page and
send a temporary
password
and instructions



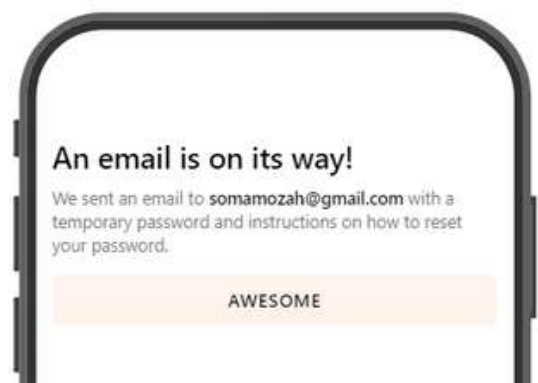
Password Reset

Enter your email address and we'll send you instructions on how to reset your password.

Email

SUBMIT

CANCEL



An email is on its way!

We sent an email to **somamozah@gmail.com** with a temporary password and instructions on how to reset your password.

AWESOME

sign up page

after pressing sign up

10:16 Philagive

Sign Up

Email

Password

Full Name

Phone number

Location

Username

SIGN UP

ALREADY HAVE AN ACCOUNT?

verification page

after pressing sign up

10:20

verify your email & phone number

a code of 4 diget sent to your email and phone number , enter the codes

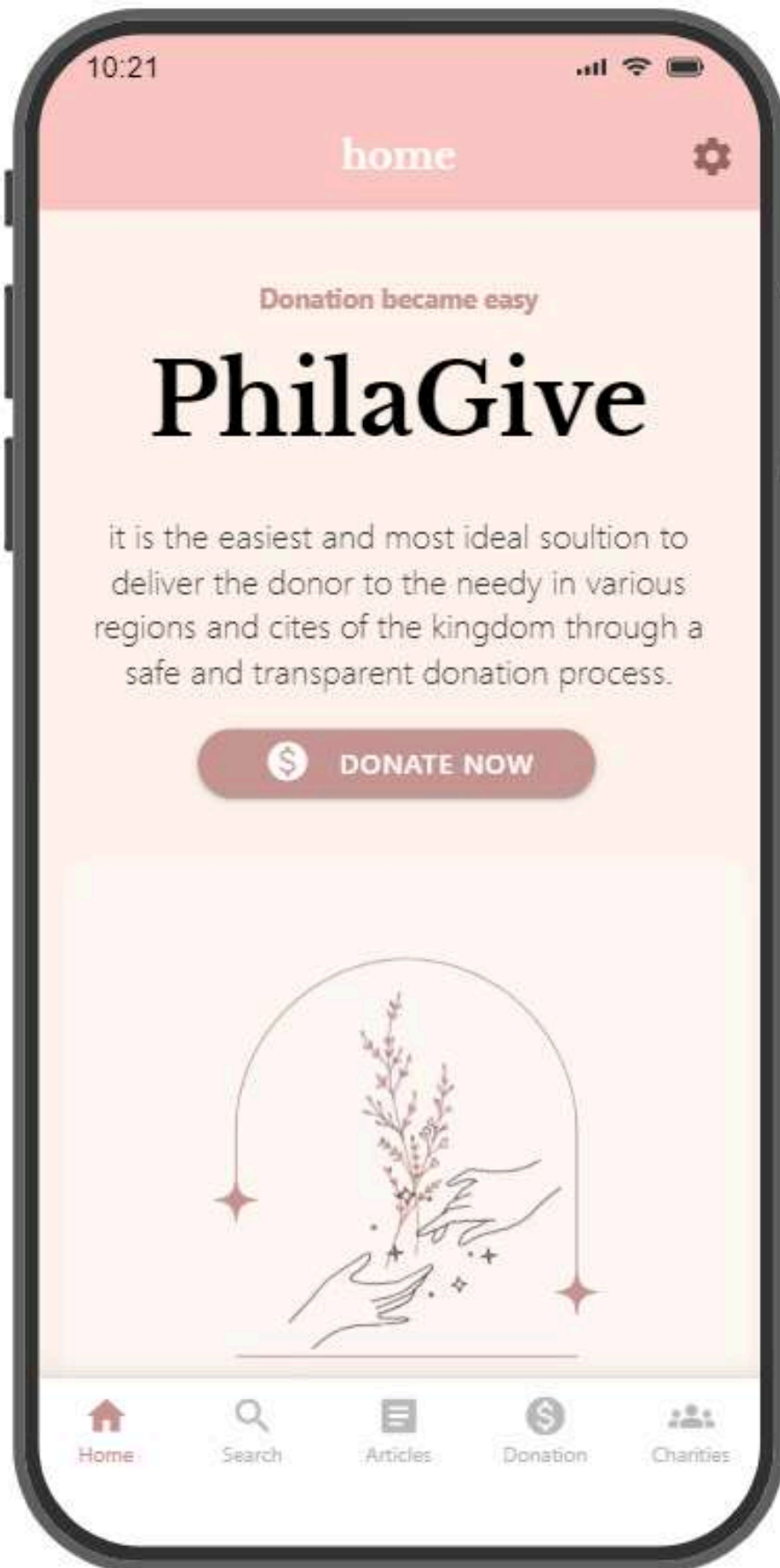
email code

phone num code

verify

home page

after verifying it will move to home page



pressing on
(setting icon)
will move the
user to setting
page

pressing on
(donate now)
will move the
user to new
donation page

moving
using the
navigation
buttons
here

setting page

pressing setting icon

the user can
update the
info
and will be
able to sign
out of the
account
and delete it

11:13

Setting

Account Settings

Name
maya Save

Email
maya@gmail.com Save

Location
makkah city Save

Phone number
506,514,523 Save

Password
Enter new password

SIGN OUT

11:15

Setting

Name
maya Save

Email
maya@gmail.com Save

Location
makkah city Save

Phone number
506,514,523 Save

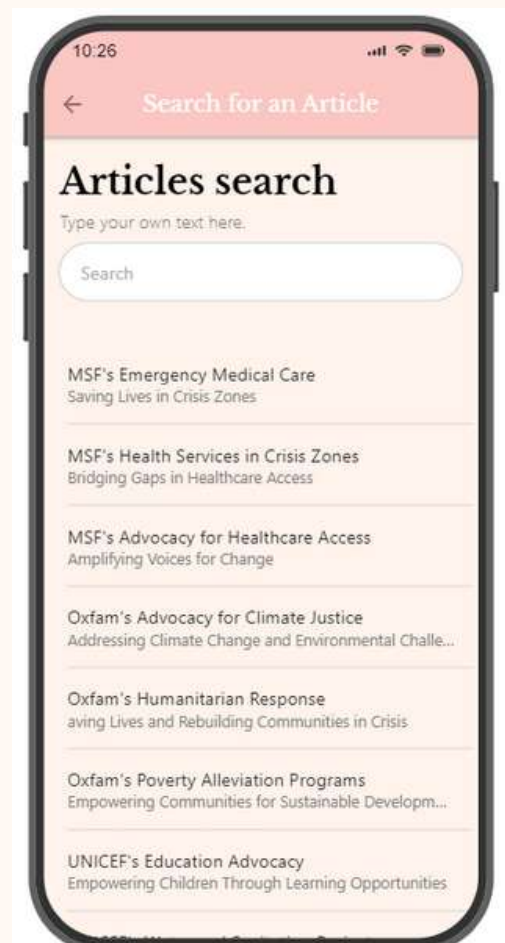
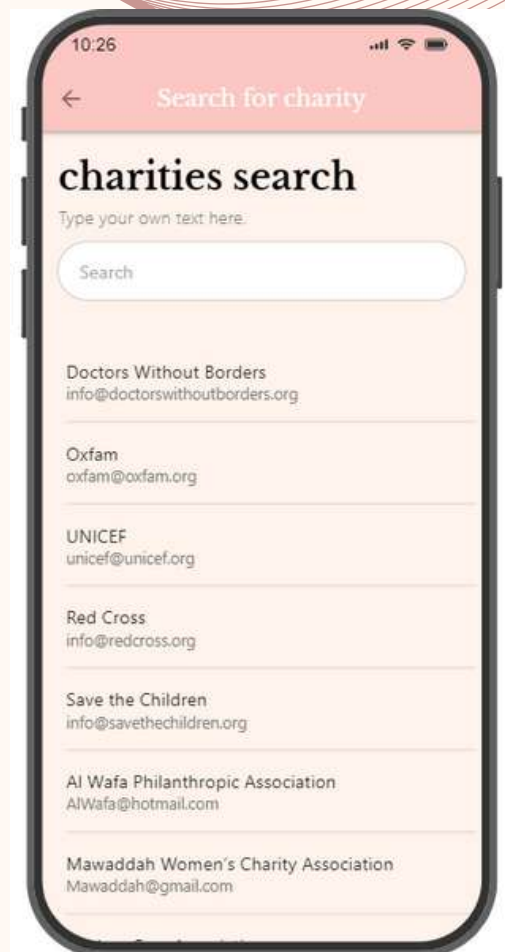
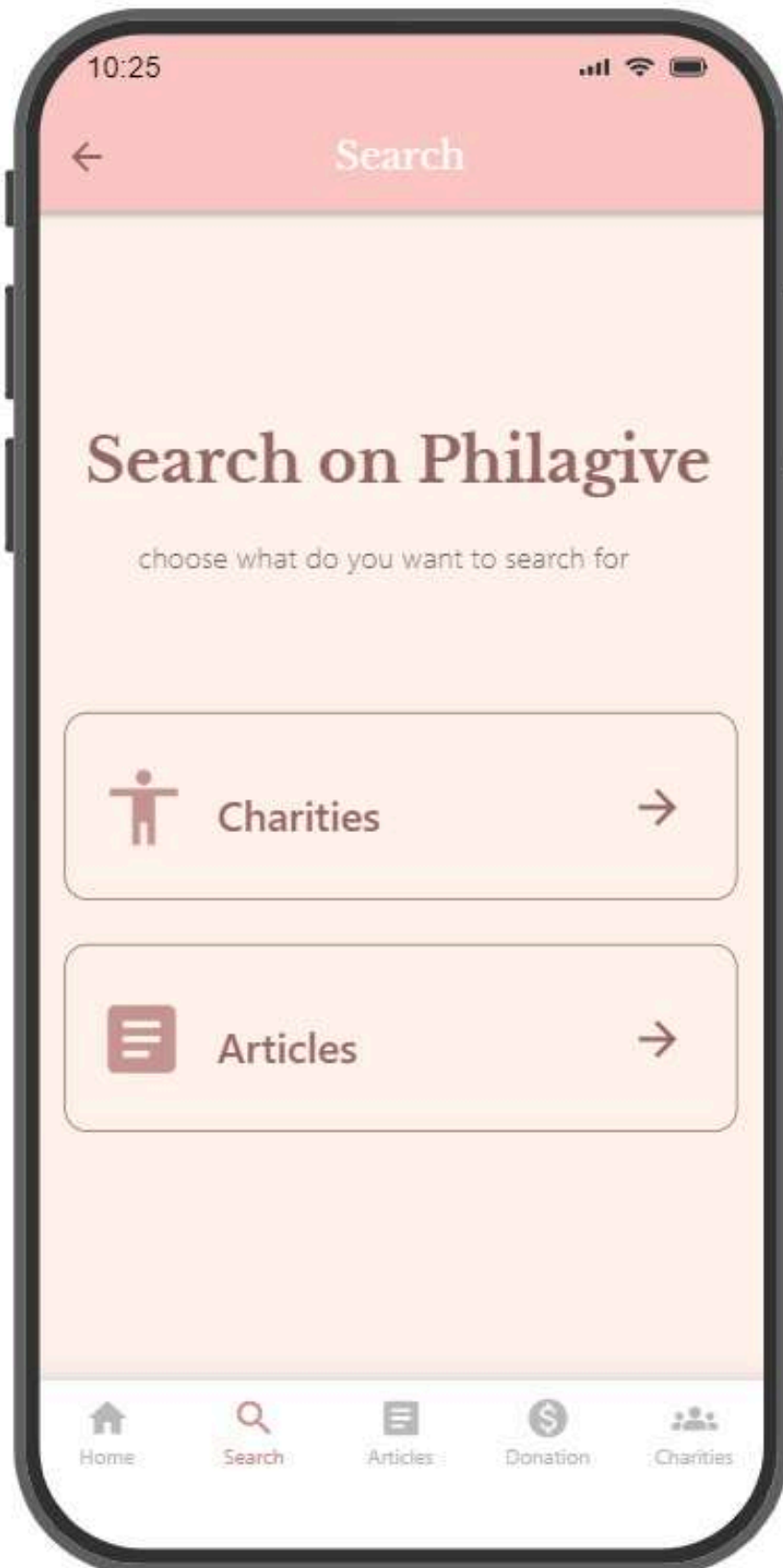
Password
Enter new password

SIGN OUT

DELETE ACCOUNT

search page

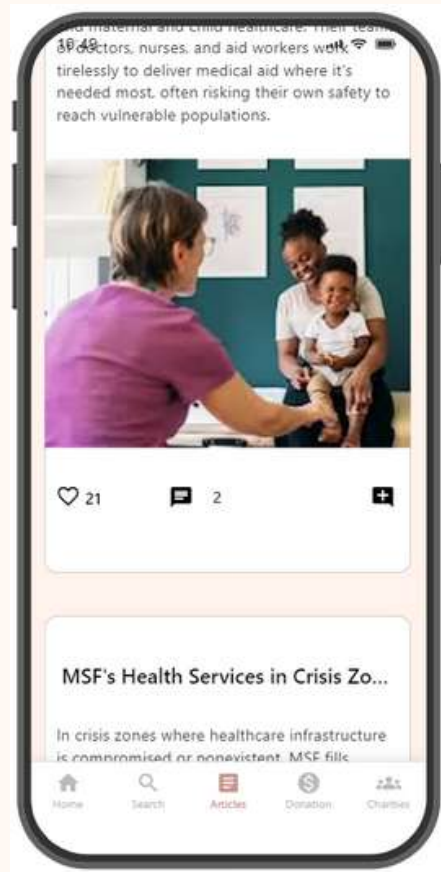
pressing on buttons will move us
and we can type and search



articles page

sarah

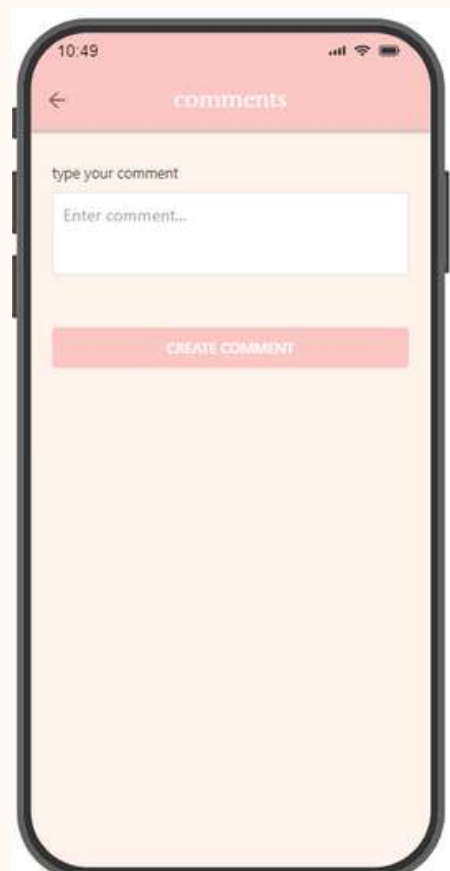
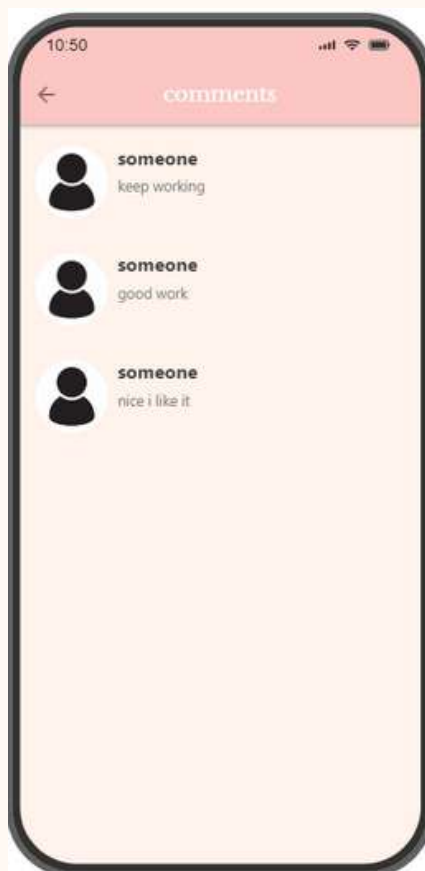
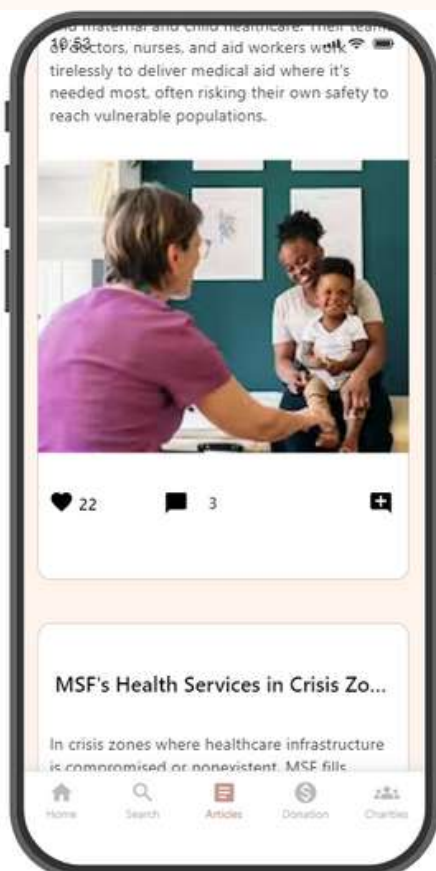
we can scroll down and see all articles
and we can react with it and add comments and view it



(like icon):
add like to article.

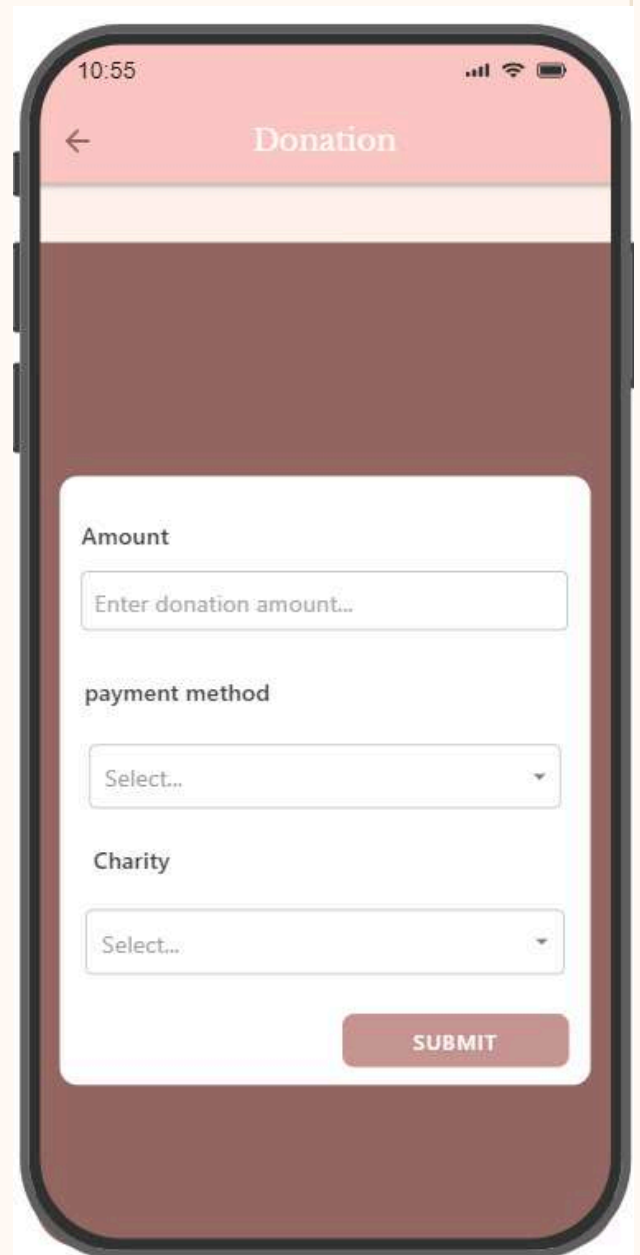
(view comments):
display all
comment of the
article.

(add comment):
display form to
create comment



Donation page

dawlah



Donation page

new donation
enter amount
select payment method
select charity

10:55

← Donation

Amount

Enter donation amount...

payment method

Select...

Charity

Select...

SUBMIT

10:55

← Donation

Amount

Enter donation amount...

payment method

Select...

Select...

mada

master card

visa

Apple pay

SUBMIT

10:56

← Donation

Select...

Doctors Without Borders

Oxfam

UNICEF

Red Cross

Save the Children

Al Wafa Philanthropic Association

Mawaddah Women's Charity Association

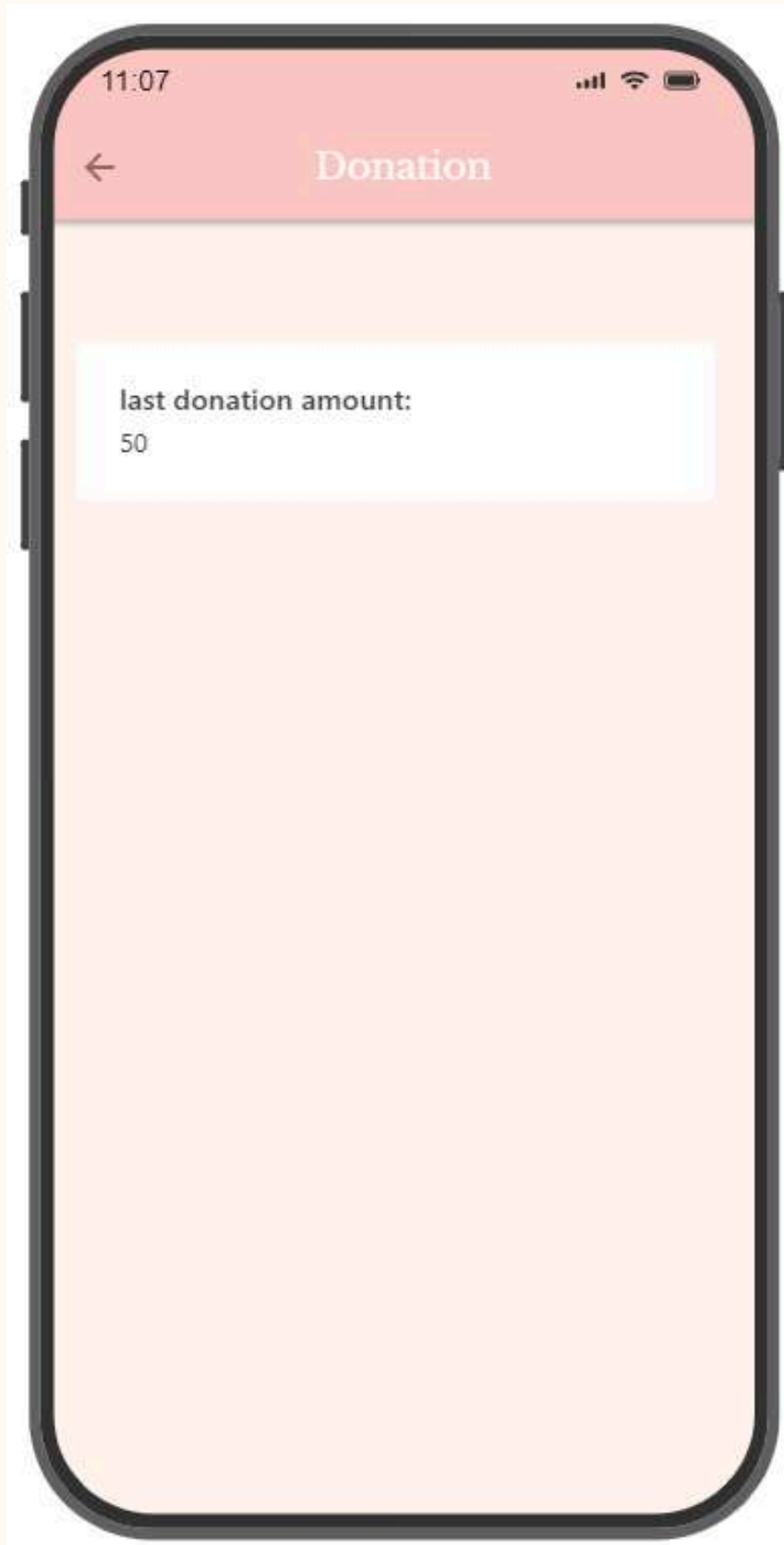
Orphan Care Association

Select...

SUBMIT

history of Donation page dawlah

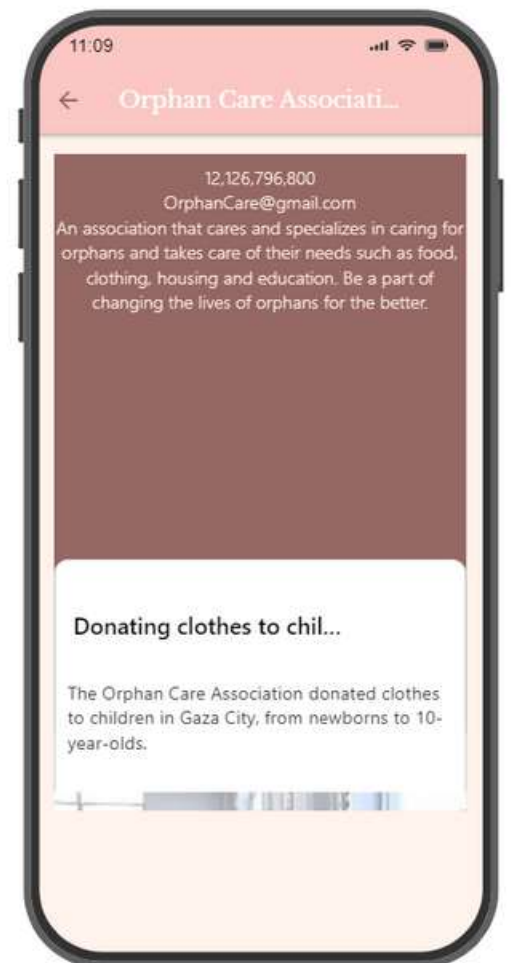
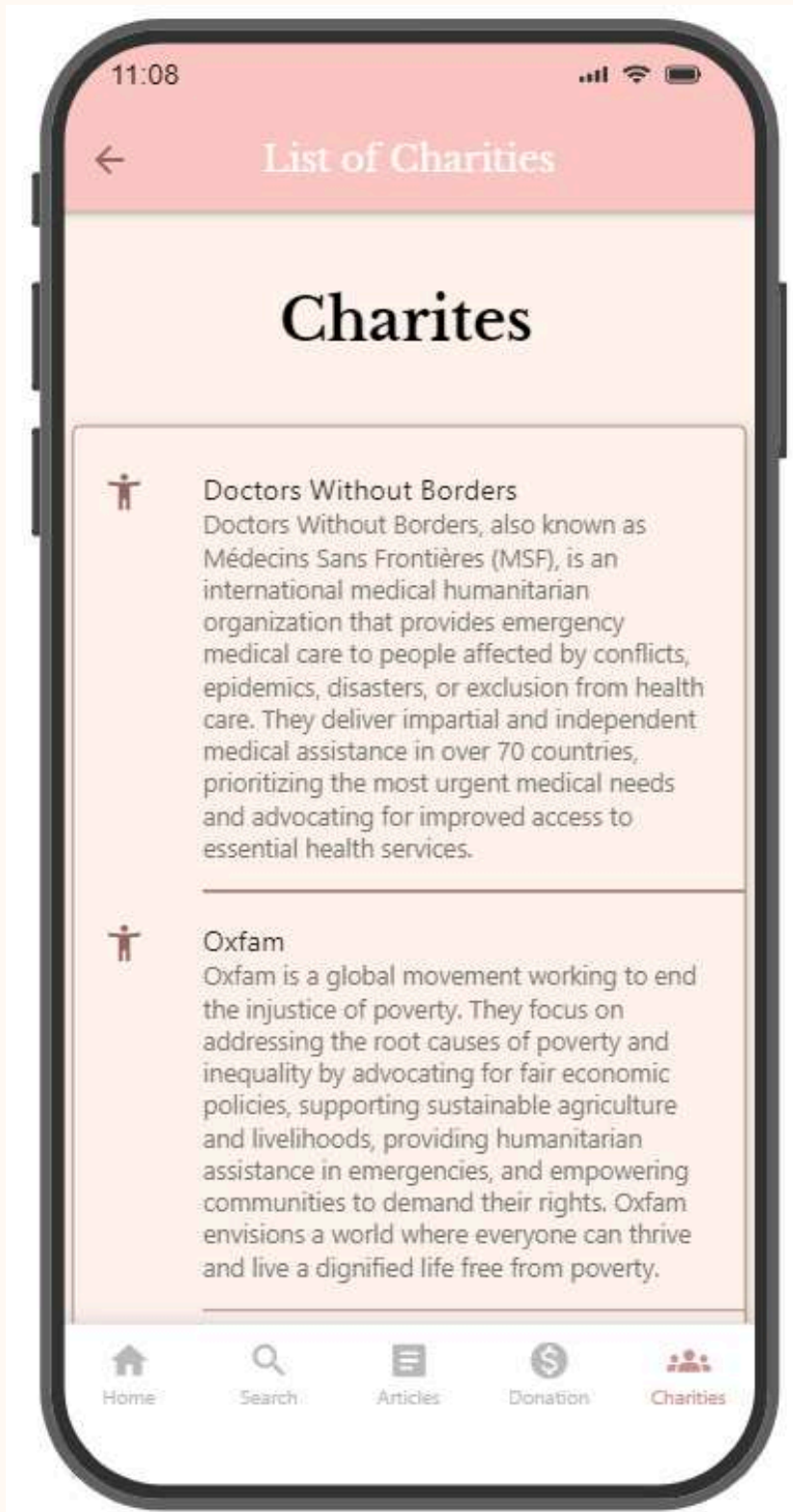
pressing on submit will update the last donation and display this page



list of charities page

maysam

list of all charities, pressing on the charity will move to the charity's page on the page, there is contact info and description and all articles about the charity



try the application!



philagive 2024

By maysam's Team

SHARE APP



Scan to Install

*If your camera doesn't have a QR code scanner,
you can [send the link via sms](#) or [copy the link here](#).



The background of the page features a series of thin, flowing red lines that create a sense of movement and depth. These lines originate from the left side and curve towards the right, forming a shape reminiscent of a stylized 'S' or a ribbon. The lines are closely spaced, creating a dense, textured effect in some areas, while other areas are more open. The overall color palette is a soft, muted red against a light beige background.

testing

Welcome page testing

ID	The case	Expected output	Actual output	Status
001	When you click the “Sign up” button, does it take you to the Sign-up page?	Move to the next page to Sign up	Move to the next page to Sign up	Pass
002	When you click the “Already have an account?” button, does it take you to the Log-in page?	Move to the next page to Log-in	Move to the next page to Log-in	Pass

log in page testing

futun

ID	Test case	Input	Expected output	Actual output	Remarks
TC-II0	The user enters the email correctly	Email saved in database	Enter the home page	Enter the home page	PASS
TC-III	The user enters the email incorrectly	The email is not saved in database	invalid Email or password	invalid Email or password	PASS
TC-II2	The user enters the password correctly	The password is saved in database	Enter the home page	Enter the home page	PASS
TC-II3	The user enters the password incorrectly	The password is not saved in database	invalid Email or password	invalid Email or password	PASS
TC-II4	The user does not enter anything	nothing	Red color on the email and password, as it requires the user to enter data	Red color on the email and password, as it requires the user to enter data	PASS

sign up page testing

sarah

ID	The case	Expected output	Actual output	Status
001	When you click the “Sign up” button, does it take you to the data registration page?	Move to the next page to sign up	Move to the next page to sign up	Pass
002	When entering data, the system notifies the user if the data is entered incorrectly	The system does not move the user to the next page and indicates to the incorrect field	The system does not move the user to the next page and indicates to the incorrect field	Pass
003	Save user info in the database	See the user info in database	See the user info in database	Pass

verification page testing

ID	The case	Expected output	Actual output	Status
001	After pressing the sign-up button the user will receive a code on his number	The user enters the code and then clicks "Verify" and the user is taken to the home page	The user enters any code and then clicks "Verify" and the user is taken to the home page	Fail
002	After pressing the sign-up button the user will receive a code on his email	The user enters the code and then clicks "Verify" and the user is taken to the home page	The user enters any code and then clicks "Verify" and the user is taken to the home page	Fail

home page testing

maysam

Test case ID	Test case description	Test case steps	Test case data	Expected results	Actual results	status
TC-01	Check how quickly the home page loads.	Launch the PhilaGive app, then go to the main page.	-	The home page loads in a reasonable amount of time.	The home page loads in a reasonable amount of time.	pass
TC-02	Verify the home page's visual layout and design.	Look for any differences in the home page's visual layout and design.	-	The home page appears as intended, with all components and text in their proper alignments.	The home page appears as intended, with all components and text in their proper alignments.	pass
TC-03	Verify that every navigation link on the main page works properly.	Select every link (button) in the navigation on the main page.	-	Every button directs users to the relevant sections or carries out the specified functions.	Every button directs users to the relevant sections or carries out the specified functions.	pass
TC-04	Check that the information on the main page is accurate and pertinent.	Examine every piece of written and visual content on the main page.	Content on the home page	The data displayed is in line with the objectives and purpose of the PhilaGive app	The data displayed is in line with the objectives and purpose of the PhilaGive app	pass

setting page testing

maysam

Test case ID	Test case description	Test case steps	Test case data	Expected results	Actual results	status
TC-01	Check to see if there are fields on the settings page for changing the location, phone number, email address, username, and password.	1. Go to the page with settings. 2. Confirm that fields for changing the username, password, name, email address, phone number, and location are present.	Details of the user account	There are all the necessary fields for updating an account.	There are all the necessary fields for updating an account.	pass
TC-02	Make sure the user can submit modifications to their account information via the settings page.	Add fresh data to the fields. 2. Press on "SAVE"	Details of a new user account	The information of the user will be updated	The information of the user will be updated	pass
TC-03	Check the settings page to see if it can continue to verify and uphold the integrity and accuracy of data in the database after updates..	1. Update user data with accurate information. 2. Check the data that has been stored in the database.	Valid user account details	Accurate information is kept in the database.	Accurate information is kept in the database.	pass
TC-04	Verify that the settings page protect private user information.	1. Examine the security precautions put in place on the settings page.	security procedures	Security procedures are followed to protect sensitive user data.	Security procedures are followed to protect sensitive user data.	pass

search page testing

futun

ID	Test case	Input	Expected output	Actual output	Remarks
TC-120	The user clicks on Search Charities button	clicks on Search Charities button	The Search Charities interface will appear	The Search Charities interface will appear	PASS
TC-121	The user enters a keyword to search for the charities he wants	The user enters (Al) in the search	Al Wafa Philanthropic Association will appear	Al Wafa Philanthropic Association will appear	PASS
TC-122	The user clicks on Search Articles button	clicks on Search Articles button	The article search interface will appear	The article search interface will appear	PASS
TC-123	The user enters a keyword to search for articles he wants to read	The user enters (care) in the article search	An article about (MSF's Emergency Medical Care) will appear	An article about (MSF's Emergency Medical Care) will appear	PASS

articles page testing

sarah

ID	The case	Expected output	Actual output	Status
001	When the user clicks on the articles button does it move him to articles page?	The user moves to the articles page and all articles are shown to him and he can browse them	The user moves to the articles page and all articles are shown to him and he can browse them	Pass
002	Can the user press the like button?	The user can press the like button	The user can press the like button	Pass
003	Can the user add a comment to the article	The user can add comment	The user can add comment	Pass
004	Can the user read the comments?	The user can read the comments	The user can read the comments	Pass

donation page testing

dawlah

ID	The case	Expected output	Actual output	Status
001	Test the New Donation button	The button takes us to a page where we fill in donation information	The button takes us to a page where we fill in donation information	Pass
002	Does the Submit button take us to the history of the donation page?	The Submit button takes us to the history of the donation page	The Submit button takes us to the history of the donation page	Pass

history of donation page testing

dawlah

ID	The case	Expected output	Actual output	Status
001	Does the Submit button take us to the history of the donation page?	The Submit button takes us to the history of the donation page	The Submit button takes us to the history of the donation page	Pass

list of charities page testing

maysam

Test case ID	Test case description	Test case steps	Test case data	Expected results	Actual results	status
TC-01	Navigation to Charities Page	Select the "Charities" area from the navigation bar.	-	The Charities page with a list of charities is displayed to the user.	The Charities page with a list of charities is displayed to the user.	Pass
TC-02	Clicking on a Charity	On the Charities list page, clicks on a charity	Selected Charity	When you click on the charity, comprehensive details, articles, and a description of the organization should appear.	When you click on the charity, comprehensive details, articles, and a description of the organization should appear.	pass
TC-03	Page Scrolling to View Additional Content	Visit the Charities page	-	There should be more content available as you scroll down the page, including articles and a description of each charity.	There should be more content available as you scroll down the page, including articles and a description of each charity.	pass
TC-04	Accessing Articles	Select a specific article of the page of charity after scrolling down	Selected article	A thorough list of articles offering more details about the charity's work should be visible when accessing the articles about it.	A thorough list of articles offering more details about the charity's work should be visible when accessing the articles about it.	pass
TC-05	Reviewing Detailed Description of the Charity	Looking at the charity page	-	When choosing a charity, a thorough description of the organization's goals, principles, and impact should be presented.	When choosing a charity, a thorough description of the organization's goals, principles, and impact should be presented.	pass
TC-06	Search Functionality Testing	Look for a particular charity using the search bar.	Search Query (name of the charity)	The user should be able to find the specific charity's page with pertinent information by typing the charity's name into the search bar.	The user should be able to find the specific charity's page with pertinent information by typing the charity's name into the search bar.	pass
TC-07	Complete Charity Information Is Visible	Explore the Charities page	-	give users comprehensive insights and options, all charities on the list should have complete information,	give users comprehensive insights and options, all charities on the list should have complete information,	pass



evaluation

Refactoring:

Refactoring in software engineering is the process of improving existing code to make it clearer and more maintainable without changing its external behavior or functionality. This includes improving names, reorganizing code, eliminating redundancies, and enhancing structure. The goals of refactoring are to make the code easier to understand and maintain, reduce errors, and sometimes improve performance.

Bad smells:

Duplicate Code:

When you have the same or very similar code in multiple places. we can use the Extract Method technique to create a single method that can be called from multiple places.

Long Method:

When a method is too long and complex. we can break down the method into smaller methods with specific functionalities (Extract Method).

Data Clumps:

When you find the same group of variables in multiple places. we can create an object to group these data together instead of passing them individually.

Switch Statements:

When switch statements are used excessively. we can use the State Pattern or Polymorphism to avoid the switch.

Speculative Generality:

When you have more complex code than necessary, you anticipate future uncertain use cases. we have to keep the code simple and extendable only when there is a real need.